



ESTIMATION OF FUEL CONSUMPTION ON KRI BIMA SUCI USING MACHINE LEARNING

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SURABAYA
2022



ESTIMASI BAHAN BAKAR PADA KAPAL KRI BIMA SUCI MENGUNAKAN *MACHINE LEARNING*

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APPROVAL SHEET

ESTIMATION OF FUEL CONSUMPTION ON KRI BIMA SUCI USING MACHINE LEARNING

BACHELOR THESIS

Submitted to fulfill one of the requirements for obtaining an Bachelor
Engineering Degree in the field of study of *Digital Marine Operation and
Maintenance*
(DMOM)

Program Study of Bachelor Engineering (S-1) of Departement Marine
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USING MACHINE LEARNING

MAHENDRA ALFATH FIRDAUS

A thesis submitted in fulfillment of the
requirements for the award of the degree of
Bachelor of Engineering
(Field of Study in English)

Faculty of Marine Technology
Institut Teknologi Sepuluh Nopember
February 2022

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"I declare that this thesis entitled *Estimation of Fuel Consumption on KRI Bima Suci using Machine Learning* is the result of my own research except as cited in the references. This thesis has not been accepted for any degree and is not concurrently submitted in the candidature of any other degree."



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ABSTRACT

There were problems in fuel on the KRI Bima Suci, such as a lack of fuel when sailing due to the estimation calculations, which were still reasonably manual. The absence of computers is also one of the causes of problems. Therefore, this study was conducted to make a solution to this problem, the estimation of fuel consumption will be carried out using machine learning. Machine learning is a part of Artificial Intelligence that can make predictions using the regression method. By using four variables, including ship speed, wind strength, sea conditions, and engine rpm, it is expected to be able to estimate fuel consumption correctly. The research in this study resulted in classification and estimation that were classified as good. For the accuracy obtained in the classification method, K Nearest Neighbor is the highest among other classification methods, with the highest accuracy rate reaching 84.95%. For the accuracy obtained in the regression method, Regression Polynomial is the best among other regression methods, with an accuracy rate of 98.99% and an average error of only 13.66. To make it easier for KRI Bima Suci to use machine learning methods, a website prototype was created to process new data that can be entered as additional data so that the classification and regression methods are better.

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ABSTRAK

Adanya permasalahan dalam bahan bakar pada KRI Bima Suci, seperti kekurangan bahan bakar saat sedang berlayar disebabkan karena perhitungan estimasi yang masih terbilang manual. Belum adanya penggunaan komputerisasi juga salah satu penyebab timbulnya permasalahan. Oleh karena itu, studi ini dilakukan untuk membuat penyelesaian dalam permasalahan tersebut, estimasi konsumsi bahan bakar akan dilakukan menggunakan machine learning. Machine learning ialah suatu bagian dari Artificial Intelligence dimana bisa melakukan suatu prediksi dengan metode regresi. Dengan menggunakan 4 variabel diantaranya kecepatan kapal, kekuatan angin, kondisi laut, dan engine rpm diharapkan bisa mengestimasi konsumsi bahan bakar dengan baik. Penelitian dalam studi ini menghasilkan suatu klasifikasi dan estimasi yang tergolong baik. Untuk akurasi yang diperoleh dalam metode klasifikasi, K Nearest Neighbor merupakan yang tertinggi diantara metode klasifikasi yang lainnya, dengan tingkat akurasi terbesar mencapai 84.95%. Untuk akurasi yang diperoleh dalam metode regresi, Polinomial Regresi merupakan yang terbaik diantara metode regresi yang lainnya, dengan tingkat akurasi sebesar 98.99% dan rata-rata error hanya 13.66. Dalam memudahkan pihak KRI Bima Suci dalam menggunakan metode machine learning, dibuatkan suatu prototipe website yang berfungsi untuk pengolahan data baru yang bisa dimasukkan sebagai data tambahan agar metode klasifikasi dan regresi bertambah baik.

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LIST OF SYMBOLS

X_n	- Independent Variable
b_n	- Coefficient or the slope of the line
b_o	- Constant or the intercept of the best-fit regression line ship speed
$x1$	- Ship speed variable
$x2$	- Wind Power variable
$x3$	- Sea condition variable
$x4$	- Engine RPM variable
y	- Fuel Consumption estimation
$\eta s \eta b$	- Losses
Pc	- Propulsive Coefficient
ηH	- Hull efficiency
ηrr	- Relative rotational efficiency
ηo	- Propulsion efficiency
Caa	- Air resistance
Ct	- Total ship resistance
Cf	- Frictional resistance
Cr	- Residual resistance
Ca	- Incremental resistance
Cas	- Steering resistance
Lpp	- Long perpendicular
Cb	- Coefficient block
B	- Breadth
T	- Draught
H	- Height

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CHAPTER 1

INTRODUCTION

1.1 Research Background

Shipping is a relatively energy-efficient, environmentally friendly, and sustainable mode of transportation (Chapman El, 2007). Therefore, many parties use ships as an alternative to other modes of transportation. Shipping isn't different from any other industry and is heavily influenced by fuel prices. (WBCSD, 2001) mentioned that fuel oil plays an important role in all modes of transportation, especially ships, which account for more than 70% of energy use. This dependence causes ships to become emitters of greenhouse gases. And the cost of using fuel is also the highest cost of other ship operational costs. However, there is usually monitoring and controlling fuel consumption with operations designed to be better than before (DNV, 2010).

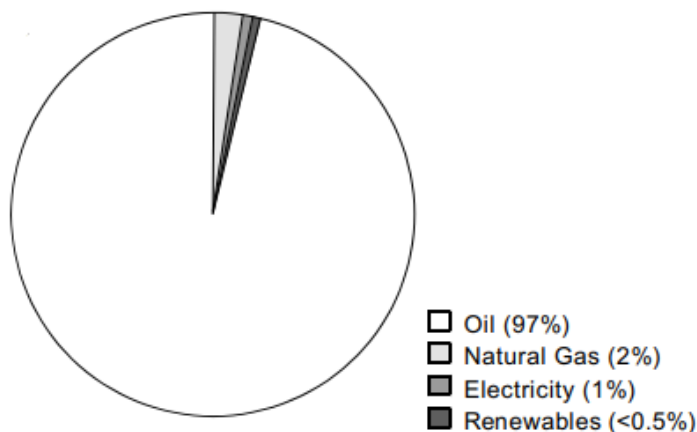


Figure 1.1 Fuel use in the transportation sector in OECD

Source (IEA, 2002)

Not only from the perspective of world trade, as from the KRI Bima Suci that are usually carried out by various parties such as the Indonesian army or other parties. Training ships can explore different types of waters and multiple countries. KRI Bima Suci is a ship that is always used by Indonesian Navy cadets to directly apply their knowledge directly in the field with various corps that have been determined. According to the Navy, KRI Bima Suci experienced a lack of fuel in 2019 when traveling to Japan and Russia. A shortage of 2-3 tons is a reasonably significant shortage. This deficiency is caused by not predicting wind and sea conditions.

In predicting the use of fuel consumption, various ship data can be processed using machine learning. Machine monitoring, ship route optimization, sailing time, and different other decisions can be managed using machine learning. With various data and analyses carried out, it is hoped that it will positively impact ship owners. In this study, the model will be made to predict fuel use on training ships so that results are obtained to reduce ship fuel use. In recent decades there have been many methods related to fuel prediction. One of the variables used is to use the ship's speed as a reference. Another way is to look at sea conditions as a reference (R Zaccone, 2018), using AIS as a supporting tool (Perez HM, 2009). In 2017 at the IEEE Conference, all of the variables discussed above will reference the machine learning method. This method has worked in several other applications.

This study uses three classification methods and three machine learning regression methods in predicting ship fuel consumption, considering all the parameters available as time series. The data used are from the ship noon report and the ship engine logbook, with various parameters such as ship speed, wind power, sea condition, and engine rpm. Fuel level is obtained from engine logbook data for the 2021 trip.

1.2 Problem Statements

The problems of the study are as follows;

1. What is the most decisive variable in fuel consumption?
2. How is the machine learning modeling?
3. What is the most effective method in estimating fuel use?

1.3 Study Objectives

The purposes of the study are as follows;

1. To find out the variables that have the most impact on ship fuel consumption
2. To know how the machine learning modeling
3. To find out the most effective method in the estimation of fuel consumption

1.4 Scopes and Limitations

The scope and limitation of this bachelor thesis are:

1. The machine learning model contained in this study uses the python programming language;
2. Data adjustments caused by privacy on the part of the Indonesian Navy

1.5 Study Contribution

The contributions of the study are as follows;

1. The machine learning model used can continue to be developed as ship data is obtained in real-time;
2. It can be used as a reference, especially by the Indonesian Navy cadets in predicting fuel consumption
3. A machine learning model that can be adjusted according to the needs of the Indonesian Navy

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CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In 2006, the Ministry of Transportation in Indonesia mentioned that ship operating costs are a combination of several ships operating relationships in a voyage, divided into several components while the vessel is anchored or sailing. Ship operational costs are divided into direct costs and indirect costs.

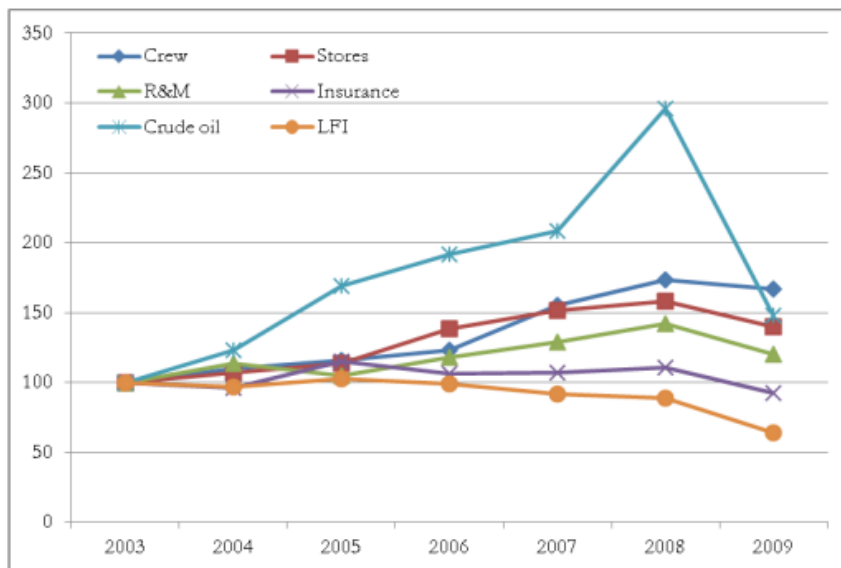


Figure 2.1 Crude oil prices, liner freight and operating costs indices from 2003-2009

Source: (M Pocuca & Zanne M, 2011)

(M Pocuca & Zanne M, 2011) also mentioned The maritime market is influenced by economic cycles. This is illustrated by changes in costs in terms of transportation, new and used, or iron and others. Ship operators will usually pay attention to these costs if a crisis hits them. The most frequent is a reduction in ship speed and thus decreased fuel costs. Although the cost of ship fuel is the highest in operating the ship, usually this cost depends on the time of the voyage and will also be affected by the price of crude oil and also the speed of the vessel itself.

2.2 Ship Fuel Consumption

(J Arnold Jr, 1991) mentioned that ship fuel consumption is one part of the ship's operating cost that consumes a lot of expenses, about 50% of the cost of the ship's operating cost is taken over by the cost of fuel consumption.

The ship voyage is usually shown about the comparison curve between the ship's power and the ship's speed. The ship's power has better stability than the ship's speed. Therefore, in terms of measuring power, it is usually more accessible. Still, the speed of the ship itself becomes the most significant parameter when determining the resistance and power of the Ship (N Bialystocki & Konovessis D, 2016).

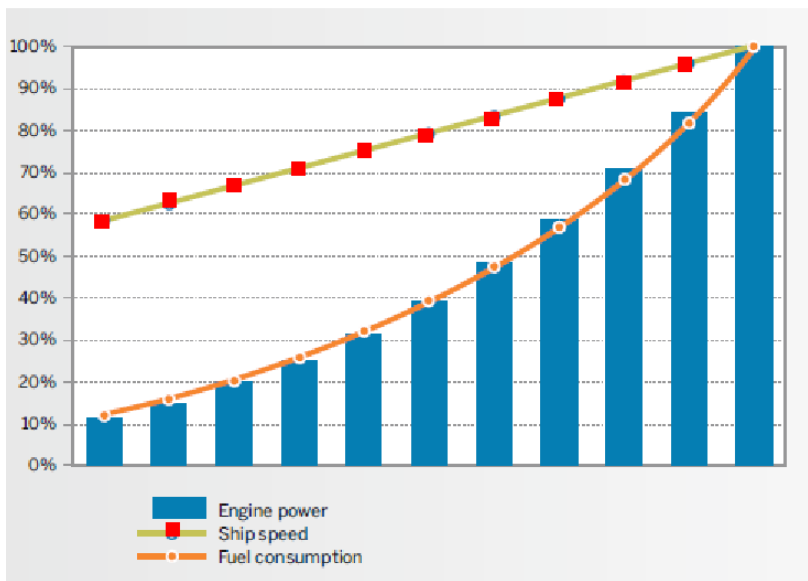


Figure 2.2 Correlation between speed and fuel consumption

Source: (Wartsilla, 2010)

Like a car or motorcycle, the faster, the heavier, the more obstacles in moving it will move linearly concerning the amount of fuel used. The ship itself has different types and characteristics. Therefore, the parameters in the calculation of fuel consumption will be different.

2.2.1 Power and Resistance

In calculating the ship's fuel consumption, several supporting factors are needed: engine load, engine rpm, fuel flow rate, and engine power (Baharuddin, 2016). So it can be concluded that the formula for ship fuel consumption is as follows:

$$SFOC = \frac{m_f + 10^3}{BP} \quad 2-1$$

In which : SFOC = Specific Fuel Oil Consumption

BP = Break Power

m_f = fuel flow rate

Formula 2-1 shows that the fuel flow rate can be obtained from the specifications of an engine. Still, the selection of ship engines depends on the calculation of ship power, and ship resistance carried out before producing a ship. Ship power has many types: Break Power, Shaft Power, Delivery Power, Thrust Power, and Effective Power (Harvald, 1983). Of the many kinds of ship power, several factors affect each type of ship power calculation.

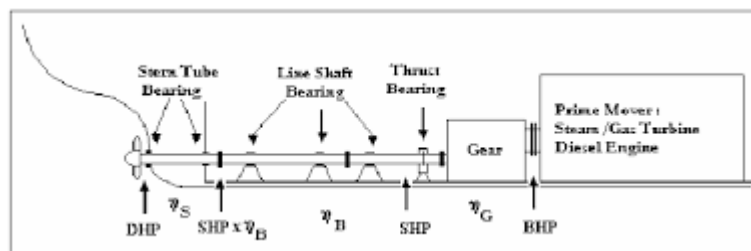


Figure 2.3 Power distribution

Brake Power is often referred to as Shaft Power because it is used for combustion engines, indicating power in a turbine or diesel engine. (Gerr, 2001). Usually, BHP will be reduced by around 3% because there is a minus internal machinery. BHP estimates specific values such as load factors and others. Equation 2-2 shows the relationship between BHP and SHP.

$$BP = SP \quad 2-2$$

Shaft Power is power that is sent to the shaft by the ship's engine, where Delivered Power produces Shaft Power reduced by friction in the bearing and stern tube. Usually, the loss in Shaft Power is reduced by about 2-3% depending on the location of the engine room of a ship (Lewis, 1988). Therefore, the Shaft Power calculation can be seen in the following 2-3 formula.

$$SP = \frac{DP}{\eta_s \eta_b} \quad 2-3$$

Before the power is continued to the shaft, power delivery was carried out previously. During this delivery, there were losses that also occurred from the effective power, which was determined at the beginning of the calculation. These losses occur due to losses between the stern tube and the tension meter, which are usually referred to as propulsive coefficients, wherein formulas 2-4 shows the calculation of Delivery Power.

$$DP = \frac{EP}{Pc}$$

2-4

$$Pc = \eta H \times \eta rr \times \eta o$$

2-5

The propulsion coefficient is caused by hull efficiency, relative rotational efficiency, and propulsion efficiency (Lewis, 1988). In calculating each of these factors, there are several parameters needed. For hull efficiency, parameters such as wake friction and thrust deduction factor are needed, which in formula 2-5 shows the formula of propulsion coefficient. For the relative efficiency of rotation and propulsion efficiency, only a constant.

$$TP = \frac{DP}{\eta H}$$

2-6

The propulsion efficient affects the calculation of thrust power, but there are several additional coefficients in its measurement in thrust power. Thrust power is the power absorbed by the propeller from the axle system or the power delivered to the propeller to be converted into thrust. Formula 2-6 shows the formula for thrust power.

$$EP = Rt \times Vs$$

2-7

The calculation of the power that must be considered in the manufacture of a ship starts from the effectiveness of the power. Where the effective power is the power required to move the ship in the water with speed (Harvald, 1983). Therefore, the ship's resistance and speed are needed to determine the ship's power. Formula 2-7 shows the calculation of the effective power.

$$Rt = Rwater + Rair$$

2-8

The ship's resistance significantly affects the calculation of the ship's power. The calculation of ship resistance is strongly influenced by several parameters, not only related to the sea but also related to air (Harvald, 1983). The shipping area of the ship is also very dependent on the final calculation of the ship's custody. Figure 2.4 shows the margin requirements for the ship's territorial waters.

North Atlantic route, eastward, 15–20% in summer and winter, respectively	
North Atlantic route, westward, 20–30% in summer and winter, respectively	(5.5.28)
Pacific route, 15–30%	
South Atlantic and Australian routes, 12–18%	
East Asiatic route, 15–20%	

Figure 2.4 Margin of ship's voyage area

Formula 2-8 shows that the total calculated resistance of the ship consists of the sum of the resistance in the water and the resistance in the air. Many parameters influence the ship's resistance to water. Several parameters also affect the air resistance.

$$R_{air} = 0.5 \times \rho_{air} \times CAA \times \text{area of foreship} \times v^2 \quad 2-9$$

$$R_{water} = 0.5 \times \rho_{seawater} \times C_t \times S \times v^2 \quad 2-10$$

Several parameters that affect the calculation of resistance to water consist of wetted surface area, ship speed, the density of seawater, and the total coefficient of resistance to friction. The parameters in the calculation of resistance to air consist of air density, coefficient of resistance to air friction, ship speed, and area of foreship (Harvald, 1983). Formulas 2-9 and 2-10 show the calculation form of the previous two-ship resistances. The ship's speed greatly affects the two resistances, which are quite essential parameters in calculating fuel consumption.

$$C_t = C_f + C_r + C_a + C_s \quad 2-11$$

$$S = 1.025 L_{pp} \times (C_b \times B + 1.7T) \quad 2-12$$

$$S = B \times (H - T) \quad 2-13$$

Suppose the formula for the ship's resistance to water and air is detailed. In that case, parameters will need to be calculated first, such as the coefficient of the ship's resistance to friction in the water, wetted surface area, and area of foreship. The four parameters must be calculated first by referring to the ship's data. Formulas 2-11 to 2-13 show the calculation form of these three parameters.

Formula 2-11 shows the calculation of the total resistance of the ship to water, where four factors are the parameters of this calculation. Frictional resistance, residual resistance, incremental resistance, and steering resistance are parameters of the ship's resistance to water. Formula 2-12 shows the calculation of the wetted surface area caused by several factors such as ship length, ship width, ship draft, and ship block coefficient. Formula 2-13 shows the calculation of the area of foreship where the width, height, and draft of the ship affect the calculation.

2.3 Fuel Monitoring

Various parties, both companies and ship owners, have been looking to reduce fuel consumption and increase efficiency. Therefore we need an accurate method in terms of fuel monitoring (A Sala, 2011). With the development of digitalization today, accurate and real-time monitoring of fuel consumption is also required to realize the term of an intelligent ship (H Lindstad, 2011).

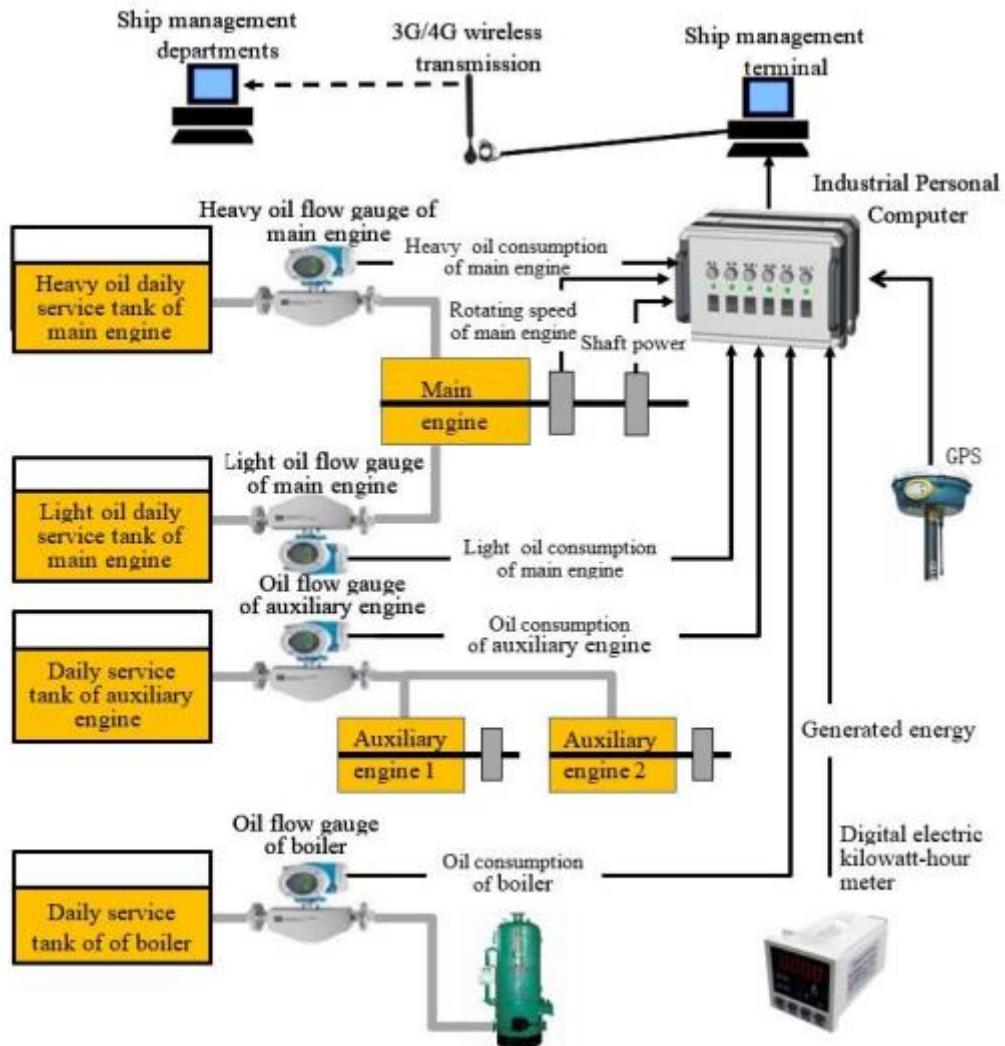


Figure 2.5 Schematic diagram of ship fuel consumption monitoring system

Source: (ICTIS Canada, 2017)

In 2017 the International Conference on Transportation Information and Safety in Canada mentioned that real-time monitoring is needed to reduce various problems in monitoring ship fuel, such as missing reports and false alarms, and can be done automatically. The measured data can be correlated to accurate measurements and detect fault alarms.

2.3.1 Telemetric System

In heavy industry, telemetry is a wireless network with a comprehensive area coverage and has an access system for inter-geographical transmission and real-time

monitoring (Kamilaris A, 2016). However, this telemetry network plays a significant role if there is remote coverage where infrastructure is minimal, and there are also rarely third parties.

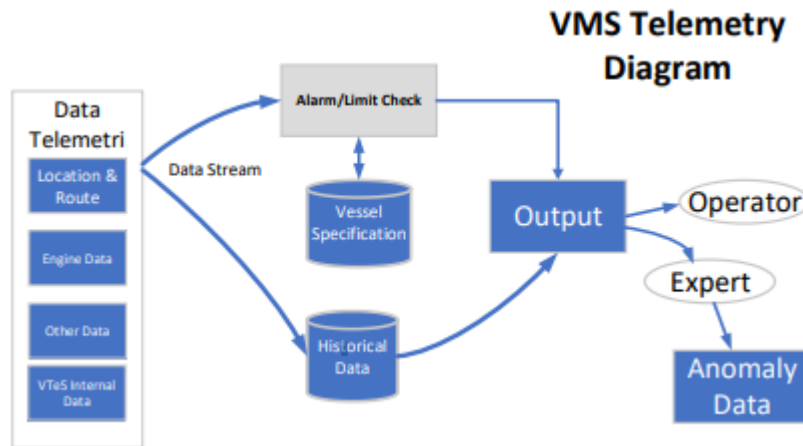


Figure 2.6 Telemetry diagram in an existing system

Source: (EECSI Bandung, 2019)

Prinsloo, Mathews, du Plessis, and Vosloo (2020) mentioned developing a telemetry system for real-time monitoring and information related to anomalies in various industries. The focus of this system is on wireless networks in a wide area to get data in real-time.

But detecting anomalies in ship movement is still possible (M Smith, 2014). There are also problems with telemetry used on ships, such as many factors that will affect fuel consumption. The high number of voyages and the number of scattered ships will affect the accuracy of the telemetry itself (EECSI Bandung, 2019).

2.4 Machine Learning

Machine learning is a branch of material from artificial intelligence, a system that can replace or imitate human behavior in solving problems, only in the form of automation and in real-time. The hallmark of machine learning itself is the existence of a training feature, which requires a lot of data that will later be classified according to the direction of the machine learning maker itself. After that, predictions will be made to guess the output of the data that has been studied and classified (Goldberg & Holland, 1988). Several categories are included in machine learning, such as supervised learning, unsupervised learning, and reinforcement learning (Somvanshi & Chavan, 2016).



Figure 2.7 Film recommended by Netflix when a consumer has watched a film

(Bell & Koren, 2007) mentioned that many applications are related to machine learning, such as the amazon bookstore and Netflix in Figure 2.7. Both applications use the decision to purchase a book or film that has been done to predict what book or film consumers will take. This machine learning system is beneficial in decision-making because it reduces guesswork and wasted time.

2.4.1 Supervised Learning

If it is interpreted logically, supervised learning is directed learning where some act as teachers and students, annotated by teachers as programmers and students as machines. The purpose of supervised learning, in general, is for classification and prediction (Brownlee, 2016). There are several classifications of supervised learning, such as binary classification, multiclass classification, and multi-label classification. Supervised learning has three things that must be considered: input, desired output, and learning parameters (Zidek, Pitel & Hosovsky, 2017).

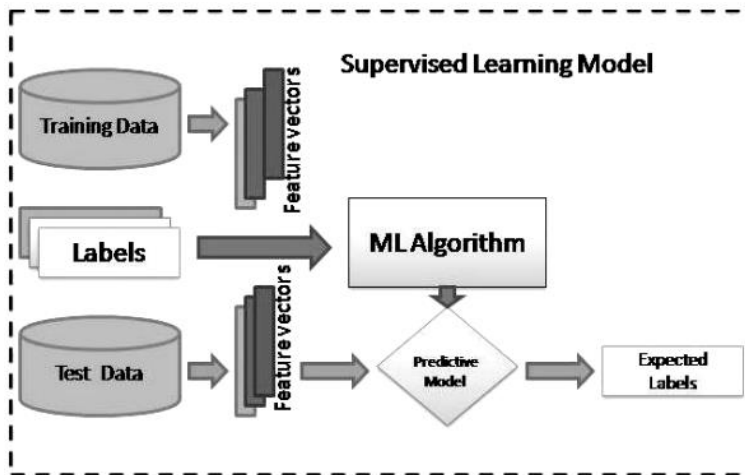


Figure 2.8 Supervised machine learning model

Source (Iqbal M & Zhu Yan, 2015)

According to (Kotsiantis 2007), the essence of supervised learning is to focus on the dataset used. In order for the data training carried out to produce an outstanding

result, a good data set is also needed. The features that exist in the tools significantly affect the dataset to be retrieved for the "brute-force" method, which is considered the pros and cons by various experts. In many cases, some feature value is missing, requiring another, more significant way.

The most frequently used use in supervised learning is classification. The data obtained will be separated into training data that serves to train the algorithm and testing data to check the feasibility of the algorithm that has been trained. Whether or not an algorithm is accepted will be determined at the end and repeated continuously to get the desired results. This repetition cycle can be carried out, depending on the validation results obtained (Osisanwo, Akinsola, Awodele, & others, 2017).

2.4.2 Unsupervised Learning

In contrast to supervised learning, unsupervised learning has no teaching aspect, so it is not directed—a learning system with sample input but no output available. The purpose of unsupervised learning itself is clustering and association. Clustering finds a group inherent in a data, while the association is a rule made to describe existing data (Somvanshi & Chavan, 2016). Problems that usually use unsupervised learning as a solution are finance, medicine, and many more (Brownlee, 2016).

(Hoffman, 2007) mentioned that this algorithm is suitable for the beginning before proceeding to the supervised learning algorithm because unsupervised grouping will identify a group that was not previously labeled with each user's provisions.

According to (Raina, Madhavan, & Andrew 2009), unsupervised learning is a very complex technique with millions of parameters and even more unlabeled data. With the help of modern graphics processors, it is hoped that this method can be carried out on very large-scale models. Currently, modern graphics processors themselves have surpassed the quality of multicore CPUs.

2.4.3 Reinforcement Learning

(Mahmud, Kaiser, Hussain, & Vassanelli, 2018) mentioned that reinforcement learning differs from supervised and unsupervised learning in that the goal is to maximize outcomes for decision sequences that can be assigned different weights. In other words, reinforcement learning does not require knowledge, so it can generate optional policies by gaining knowledge from trial and error and continuously adapting to the surrounding environment.

Usually, this method uses two primary algorithm sources, such as SARSA, which will get an output from the previous model learning, or the Q-Learning algorithm, which will get the output directly without studying an earlier model (Qiang & Zhongli, 2011). The reinforcement learning method has solved many problems in various fields, such as IoT and smart cities (Mohammadi, Al-Fuqaha, Guizani, & Oh, 2018). There is

also in the field of technology, namely computer vision (Z. Wu, Khan, Gao, & Guan, 2018).

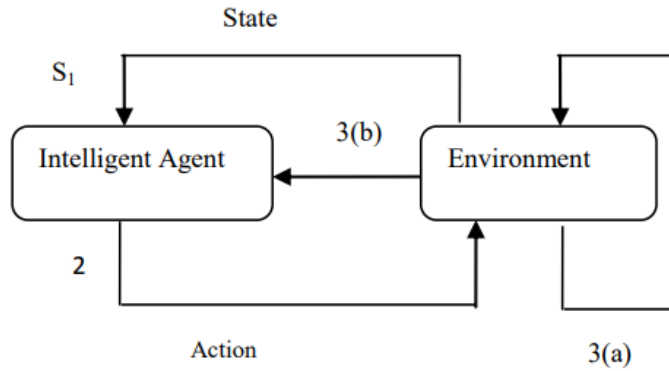


Figure 2.9 Basic model of reinforcement learning

Figure 2.7 shows that there are several primary stages in the basic model of this method, where the intelligent agent will receive input, which is an indication of a predetermined state. The intelligent agent will choose an action to produce an output. 3(a) indicates to change state, and 3(b) will transition state by communicating it via a scalar signal (Sharma, Prateek, & Sinha, 2013).

2.5 Classification and Clustering of Machine Learning

2.5.1 Naïve Bayes

(Siradjuddin, 2014) mentioned that the Naive Bayes algorithm is a technique for predicting simple probabilities according to the theorem of a British scientist named Thomas Bayes. With independent assumptions and predict future opportunities based on experience in previous situations and conditions. The equation of this algorithm can be seen below.

$$P(H|X) = \frac{P(X|H) \cdot P(H)}{P(H)} \quad (2.1)$$

The basic idea of this algorithm is that a hypothesis can be estimated based on existing evidence. According to Jhon Wiley and Sons in 2005, to explain the theorem above, the classification process requires a guide to determine the appropriate classification for the analysis sample, so some adjustments in the above equation can be seen below.

$$P(C|F1 \dots Fn) = \frac{P(C) \cdot P(F1 \dots Fn|C)}{P(F1 \dots Fn)} \quad (2.2)$$

Variable C represents the class, and F represents the characteristics of the clue. There are two stages in the calculation of the Naive Bayes algorithm, namely, Learning and Classify. (Bramer, 2007) mentioned that from the formulas above, it can be seen that this method comes from a branch of mathematics that is closely related to probability in finding the most significant opportunity. This method is also popular in various circles and includes the best algorithm—many other terms from this method such as Idiot's Bayes, Simple Bayes, and Independence Bayes.

2.5.2 K - Means

The goal in unsupervised learning is to do clustering. Where k - means is one of the basic algorithms used to perform clustering and can partition data into two or more groups (Harding, Shahbaz, Srinivas, & Kusiak, 2006). The advantages of K - Means itself are simplicity and good efficiency and is recognized as one of the many algorithms by the IEEE (X Wu, 2008).

In this algorithm, several approaches are taken, including making rules based on the level of equality or creating a function by measuring several aspects that become parameters in a clustering. Behind the positive side of this algorithm, there is also a negative side, such as, if the data obtained is small, it will be difficult to determine the initial cluster; the number of clusters must be determined before calculation. The way to overcome the negative side of this algorithm is to obtain a lot of data.

This algorithm works in two processes: looking for the center of the cluster and the cluster members. This clustering process begins with identifying the existing data, determining the amount of data to be clustered, and the number of variables. The clustering center is determined arbitrarily, then the distance of each data to each center that has been previously determined will be calculated. Data will enter a specific center if it has the smallest distance compared to the distance to other centers (Buslim, 2019).

2.5.3 K – Nearest Neighbor

The K Nearest Neighbor algorithm is one of the Lazy Learning algorithm techniques and one of the decision-making methods with supervised learning techniques which aims to classify new objects based on attributes and samples of data that have been trained (Wu, 2009).

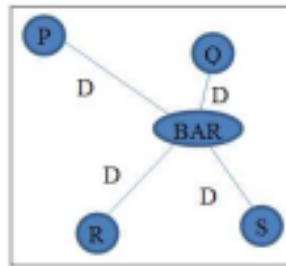


Figure 2.10 k Nearest Neighbor Illustration

Source: (Leidiyana H, 2013)

In using this algorithm, the first thing to do is look for K, where k is a parameter that must be present in performing calculations using this algorithm. In determining the value of K, it is better to use odd values. The goal is that the possibility of getting an answer is more significant than using actual values (Asahar T, 2016). This method will determine the data by making the data around the closest distance as a reference (Avelita B, 2013). This proximity is defined in terms of metric distances, such as the Euclidean distance (Jiawei Han, 2012).

2.5.4 Confusion Matrix

According to (Visa, Ramsey, Ralescu Knaap, 2011), good accuracy is needed in classification, especially if it has extensive data distribution. The more training data used in the classification, the better the accuracy obtained. There are two search strategies in the classification: forward selection and backward elimination. The fundamental difference between the two strategies is attributes, where forward selection focuses on adding attributes at each stage, while backward elimination will remove unused attributes.

Table 2.1 Confusion matrix in classification

		Actual Values	
		Positive	Negative
Predicted Values	Positive	TP	FP
	Negative	FN	TN
Accuracy		A	

Table 2.1 shows that four results are the confusion matrix's outputs. TP (True Positive) and TN (True Negative) are the correct classifications, while FP (False Positive) and FN (False Negative) are types of calculation errors. From these four

results, an A (Accuracy) will be found from the classification that has been carried out. The accuracy obtained can be seen in this formula.

$$A = \frac{TP + TN}{TP + FP + FN + TN}$$

2.6 Regression of Machine Learning

2.6.1 Linear & Multilinear Regression

Analysis of regression is a technique that analyzes the relationship between the dependent and independent variables. This linear regression takes one dependent variable and one independent variable. However, multilinear regression takes more than one independent variable.

According to (Kumari K & Yadav S, 2018), linear regression is used to calculate the value of the existing, independent, and dependent variables. This method measures the relationship between the two variables, where the dependent variable can be more than one. This analysis method is one that various parties often use.

$$Y = b_0 + bX_1 + e$$

In which : Y = Dependent variable

X = independent variable

b = Coefficient or the slope of the line

b_0 = constant or the intercept of the best-fit regression line.

(Ali Fikri, 2013) mentioned that there is a case with more than one independent variable in regression, which has more complex data but produces more accurate results than linear regression. Multilinear regression has the following model:

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n + e$$

In which : Y = Dependent variable

X_n = independent variable

b_n = Coefficient or the slope of the line

b_0 = constant or the intercept of the best-fit regression line.

The use of this method will produce a formula that can be used as a calculation reference to determine an estimate of the required data. Usually, the initial data obtained is still in the form of an ordinary series, which later needs to be analyzed further. Data analysis such as eliminating or replacing the average of the empty data in

order not to affect the calculation results. After that, the data will be divided into testing and training data, aiming to divide calculation data and checking data (Pramitarini, 2013).

2.6.2 Polynomial Regression

According to (Bates & Watts, 2007), if linear regression does not get accurate results or is not as expected, you can try polynomial regression, which is not linear but quadratic. In other words, polynomial models can be used in those situations where the relationship between study and explanatory variables is curvilinear. For this regression modeling, it can be seen as follows:

$$Y = b_0 + b_1X + \dots + b_nX^n + e$$

In which : Y = Dependent variable

X^n = independent variable

b_n = Coefficient or the slope of the line

b_0 = constant or the intercept of the best-fit regression line.

Sometimes, polynomials can also model a nonlinear relationship in a small range of explanatory variables. Polynomial models are compelling in handling nonlinearity because polynomials can approximate continuous functions within any given precision. (Pramitarini, 2013) mentioned that usually, the initial data obtained is still in the form of an ordinary series, which later needs to be analyzed further. Data analysis such as eliminating or replacing the average of the empty data in order not to affect the calculation results. After that, the data will be divided into testing and training data, aiming to divide calculation data and checking data.

The results of this method are in the form of formulas and additional information in the form of evaluations. The evaluation of this algorithm is mean absolute percentage error (Makridakis, Wheelright, & Hyndman, 1998) and R-squared. MAPE itself is the average percentage of error against predictive data and actual data usually, if it is below 10%, it can be declared good (Lewis CD, 1982). R-squared is the level of accuracy of the calculation algorithm made usually a number of 0.8 and above can be said to be a good category (Ostertagova, 2011). These two evaluations serve as a reference for the correction or development of this method.

2.6.3 Support Vector Regression

(Basak, Pal, & Natranabis, 2007) mentioned that Support Vector Regression is part of the Support Vector Machine. SVM is a learning technique to generalize data by applying a line to minimize risk. SVR formulates a problem by finding the narrowest

area centered on a surface. The distance between the center and the outermost point is an error that can occur in calculations. For more details, see the illustration below:

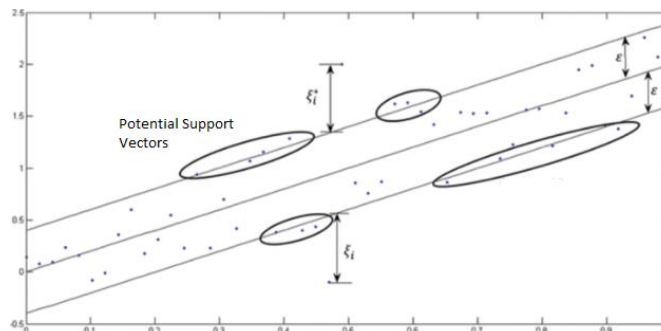


Figure 2.11 SVR Illustration

Source: (Awad & Khanna, 2015)

Figure 2.9 shows that the purpose of SVR is to try to combine existing data by minimizing prediction errors. The farthest distance of the hyperplane to the center point is usually called the margin, so maximizing the margin will increase the probability of the data within the radius and make the function as thin as possible. The resulting error is not significant (Awad & Khanna, 2015).

In its application, Support Vector Regression can overcome problems such as over-fitting. Therefore, this method produces good performance because it will minimize errors in the calculation (Ibrahim & Wibowo, 2014). Not only that, but this method also concludes about its advantages and accuracy of calculations (De Paz, Pelez, Gonzales & Chorcado, 2010). This method can also be applied for continuous data, as in 2004 by Smola & Scholkopf.

2.7 Programming and Website Development

2.7.1 Python

Python is a versatile programming language that Guido Van Rossum first created. This language is designed to make code easy for others to read. Python is portable like most other programming languages, and memory management functions as a garbage collection method in the Java programming language. Python is also an object-oriented programming language, where the paradigm is to combine properties and behavior in one object (Abdul Kadir, 2019)

For a computer scientist, it takes at least one programming language. The chosen programming language is based on innovation and technology, operating systems, software, and hardware. By becoming an expert in a programming language, a computer scientist will have a different way of thinking (Sanders & Langford, 2008).

The advantage of choosing Python as the selected programming language is that it can speed up the learning process (Bogdanchikov, Zhaparov & Suliyeu, 2013).

2.7.2 R

The R language was first launched in the 1990s when Ross Ihaka and Robert Gentleman were students at the University of Auckland, New Zealand. They created a statistical programming language by combining the S syntax and provided various features. In the mid-1990s, the development of the R language began, and a team was formalized as the R Core Team.

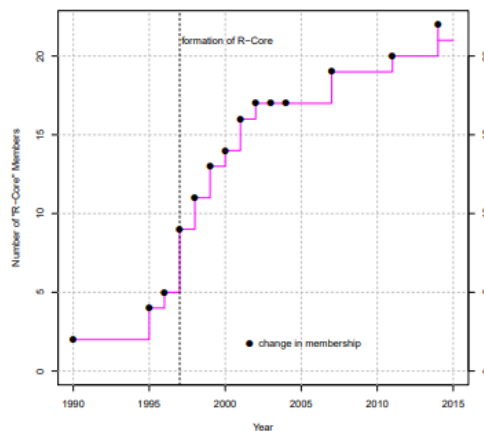


Figure 2.12 Growth in the R Core Team

Source: (Fox, 2009)

Figure 2.10 shows the growth of the R Core Team over time (Fox, 2009). It can be seen that in its development, some are fast and some are slow. According to (Ramasubramanian & Singh, 2019), the R language has several arithmetic operations like a scientific calculator. This language is easy enough for a beginner in machine learning. There is also a package that contains many functions, where the function has several algorithms needed in machine learning.

2.7.3 Figma

Figma is a cloud-based design application and prototyping tool for digital projects. The focus of this Figma application is for User Interface (UI) and User Experience (UX) Designers because this application has several tools that can simplify the process. This Figma can collaborate because it is based online, making it easier for us to work with the group online. This means that we can do online and real-time editing with our Design Team (Mailoi, 2018).

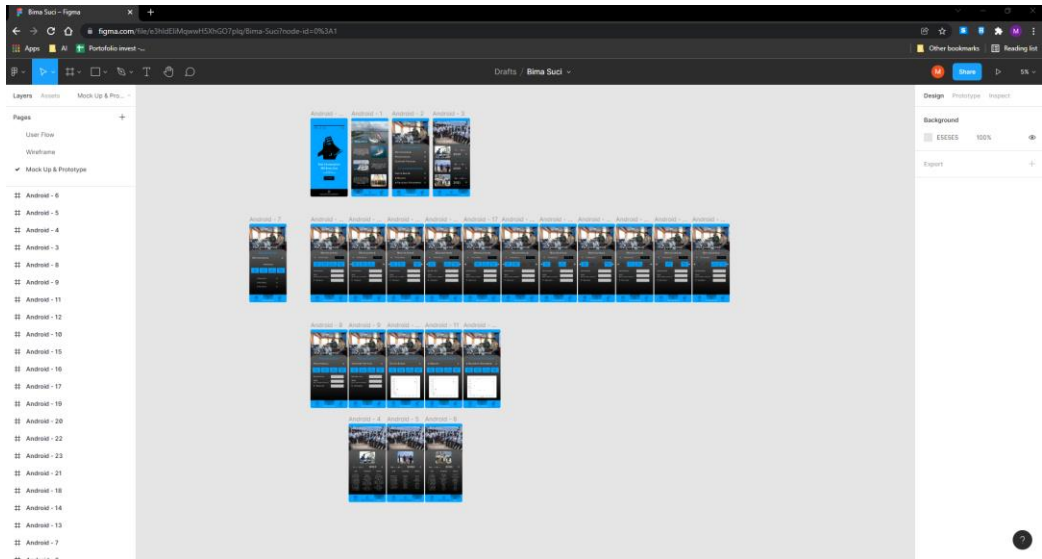


Figure 2.13 Figma platform view

Figma will talk about the ease and efficiency of interaction between users and computers through User Experience (UX). UX has the primary goal of providing comfort and satisfaction to the user. (Marsh, 2016) mentioned that there are five ingredients in UX: psychology, usability, design, copywriting, and analysis.

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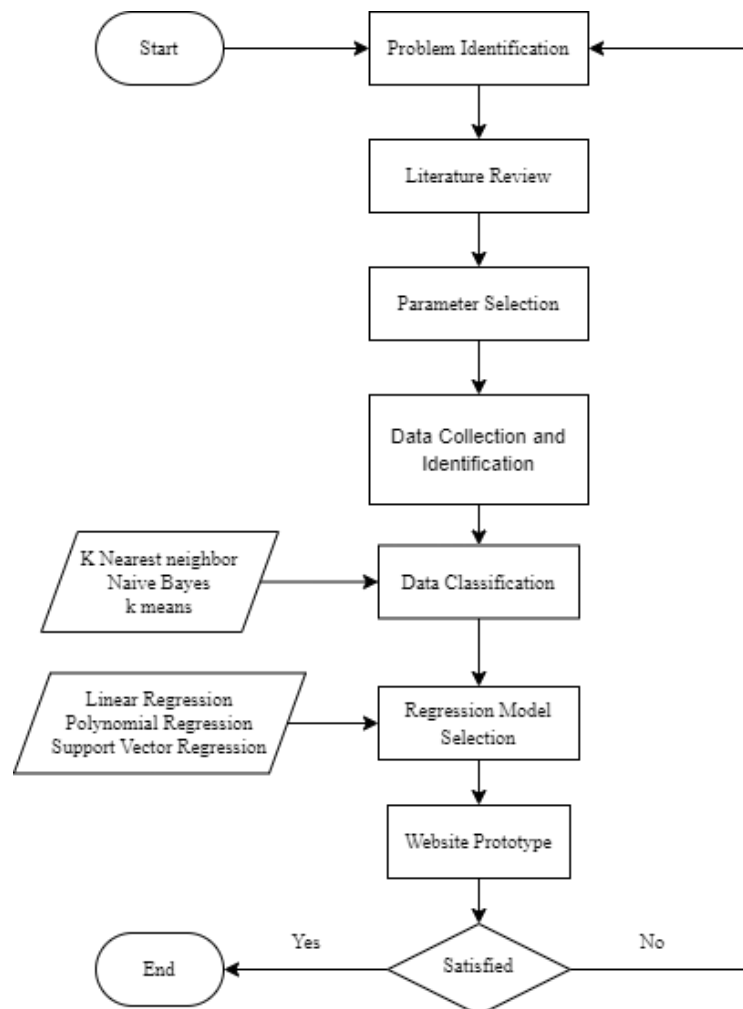
CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This study aims to estimate fuel consumption on the KRI Bima Suci using a machine learning technique and make a website easy for users to use. Study literature is needed to support the achievement of the objectives of this study. The data that has been collected will be processed through the existing system modeling. The current models are fuel consumption predictions and screen usage predictions.

3.2 Research Flowchart



3.3 Parameter Selection

Ship fuel consumption has several factors that affect the nominal change. This parameter is obtained by narrowing the existing formula. The parameters received start from those that significantly influence and those that do not affect both internal and external factors (Lewis A, 1988). In selecting these parameters, several parameters that will be used in this study were determined depending on the availability of data on the KRI Bima Suci. After determining the parameters to be used, data collection and identification will be carried out to facilitate the calculations in this study.

3.4 Data Collection & Identification

In this study, the data input and identification method uses statistical methods to collect, summarize, analyze, and analogize variables in numerical data (Kabir, 2016). The data collected is KRI Bima Suci data in ship noon reports and ship engine logbooks. Both of these data contain the results of monitoring several parameters, which later statistical methods will play an important role in the estimation of specific parameters. After estimating the parameters, direct observations will be made of the data that will be needed. The data obtained from the two sources will select specific parameters as only required in this study. The results of inputting data in historical data on ship fuel consumption and several other supporting parameters will be presented in a table to facilitate further calculations.

3.5 Data Classification

Data identification is necessary to simplify the calculation. The data obtained are several data with several variables. Data classification is used to divide the data used in the regression method and those that will not. This classification method uses machine learning to know the accuracy of the data sharing that has been done manually. The algorithm of 3 different methods is expected to get different accuracy results to make more effective comparisons. By setting the accuracy that must exceed a specific number, the data will be forwarded to be used in the regression method. If it has not passed the set value, it will be reclassified.

3.6 Regression Modeling

The data that has been classified is used in the calculations for fuel estimation. By using data that has been previously classified, the regression method will be chosen to make estimates. The three models selected include linear regression, polynomial regression, and support vector regression. In regression modeling, it will be divided into several testing data and training data. The results obtained from regression modeling are an assessment of the comparison between the results obtained with the actual results. Accuracy and error will also support calculations using this regression method.

3.7 Website Prototype

The results of the data from the regression method will compare to the actual data. This aims to see how big the difference is when using the regression system as a calculation. Accuracy and error values were also used to see how effective the regression method was in its calculations. The data obtained will make it easier for the user to assess and analyze. Therefore, we need a supporting platform that the user in processing the data can use. The prototype website was created to make it easier for users to process historical data on a ship from a specific year's trip to be used in its calculations in fuel estimation for future purposes.

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CHAPTER 4

RESULTS AND DISCUSSION

4.1 Parameter Selection

This study uses four parameters: ship speed, wind power, sea condition, and engine rpm. The reason for using 4 is that several parameters are difficult to obtain, so they cannot be used in this study, such as displacement, freshwater, and engine load.

4.1.1 Ship Speed

The speed of the ship is very influential on fuel consumption, where initially, the speed will affect the resistance quadratic. The resistance will affect the load, later affecting the machine selection. In the engine selection, it will be determined how much fuel consumption is needed.

4.1.2 Wind Power

Usually calculated on the Beaufort scale, where the wind itself plays a vital role for a ship in fuel consumption, with different strengths and different directions, it will affect the fuel consumption of a ship.

4.1.3 Sea Condition

Conditions passed by ships must be different. Calm sea conditions will have a different impact on choppy sea conditions. Therefore, sea conditions will also affect the ship's fuel consumption.

4.1.4 Engine RPM

The choice of the engine affects fuel consumption. Although engine rpm does not play a significant role in terms of fuel consumption, engine rpm will affect the engine load used. Therefore, the engine rpm will indirectly affect the ship's fuel consumption.

4.2 Data Input and Identification

The data used in this study was collected online from the ship obtained from the head of the ship's engine room, Mayor Laut (T) Budi Luhur. Data was obtained from November 16 to November 25, 2021. This research was conducted on the KRI Bima Suci training ship, with a length of 111.2 m, a width of 13.65m, and a tonnage of 2500 tons. The ship is powered by one main thruster MAN 6L21/31 engine, with a power

capacity of 1290 kW and a rotation speed of 1000 rpm. The data taken is the data on the ship's journey that was carried out in 2021, where the first training ship KRI Bima Suci was used by the Indonesian Navy Academy. The data to be used in the form of ship speed, wind power, sea conditions, engine RPM, and fuel consumption.

Table 4.1 Data example of KRI Bima Suci

Ship Speed	Wind Power	Sea Condition	Engine RPM	Fuel Consumption
9.75	1	0.75	952	1500
7.00	2	0.75	939	1500
9.43	3	0.75	898	2298
NaN	3	0.75	NaN	1100
NaN	2	0.75	NaN	1500
NaN	2	0.75	NaN	1034
NaN	1	0.75	NaN	1006
11.18	3	0.75	937	1400
11.58	3	0.75	936	1400
8.45	3	0.75	940	1400
8.80	2	0.75	915	1400
9.40	2	0.75	935	1400

There are several disclaimers in the data that I got. Attachment 1 shows examples of data collection sources, namely ship noon reports and ship engine logbooks. Table 4.1 shows that some data do not make sense, and theoretically, the speed of the ship should have a polynomial effect on fuel consumption, but in the data, I got some that were not related to theory.

4.3 Data Classification

In this classification, the data obtained will be sorted manually. The data that has been sorted will be assessed for accuracy using machine learning methods, wherein the sorted data is determined which data will be used in the regression method and which will not be used. The purpose of this classification is that the regression method can produce high accuracy, and unused data can be used as a reference by the Indonesian Navy to develop data records. The classification used here consists of k Nearest Neighbor, Naive Bayes, and k Means methods.

The numbers in the confusion matrix table and the accuracy obtained are algorithmic calculations from the K Nearest Neighbor & Naïve Bayes method. Attachments 2 and 3 describe a more detailed flow in this method.

4.3.1 K Nearest Neighbor

K Nearest Neighbor is a classification method in machine learning that is dependent. It will involve existing data even though they are not related to each other and non-parametric. Calculation of accuracy and table indications have been described in table 2.1 previously.

4.3.1.1 4 Variables (Ship Speed, Wind Power, Sea Condition, Engine RPM)

Four variables will be used as data parameters, plus fuel consumption data as well as the X-axis, and for the Y-axis is the classification decision that the author has made.

Table 4.2 Confusion Matrix of 4 variables KNN classification

		Actual Values	
		Accepted	Not Accepted
Predicted Values	Accepted	32	12
	Not Accepted	2	47
Accuracy		0.8494623655913979	

Table 4.2 shows that the value of the confusion matrix is quite good for the four variables used as parameters. The confusion matrix aims to measure the performance of a classification method. Because the accuracy obtained is quite good, the data that has been sorted will be used further at the stage of the regression method.

4.3.2 Naïve Bayes

Naïve Bayes is a classification method in machine learning that is independent, which involves data that must be related to each other and is non-parametric. Calculation of accuracy and table indications have been described in table 2.1 previously.

4.3.2.1 4 Variables (Ship Speed, Wind Power, Sea Condition, Engine RPM)

Four variables will be used as data parameters, plus fuel consumption data as well as the X-axis, and for the Y-axis is the classification decision that the author has made.

Table 4.3 Confusion Matrix of 4 variables Naive Bayes classification

		Actual Values	
		Accepted	Not Accepted
Predicted Values	Accepted	31	13
	Not Accepted	6	43
Accuracy		0.7956989247311828	

Table 4.3 shows that the value of the confusion matrix is quite good for the four variables used as parameters. The confusion matrix aims to measure the performance of a classification method. Because the accuracy obtained is good, but when compared with the k Nearest Neighbor method, the method used to continue at the regression stage is to use the k Nearest Neighbor method.

4.3.3 K Means

K Means is a machine learning method included in unsupervised learning. This method will automatically cluster existing data; therefore, humans do not have to sort the data first. Attachment 4 describes the detailed flow in this method.

4.3.3.1 Ship Speed

In this method, the ship's speed is used as a reference which will later be classified with specific parameters of fuel consumption that have been obtained.

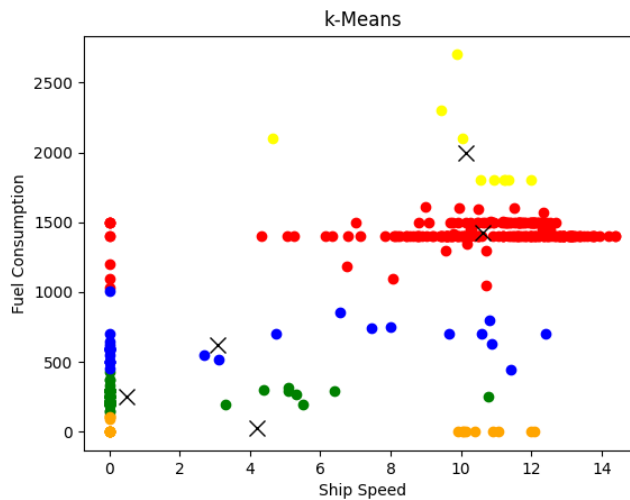


Figure 4.1 Ship Speed variable output from k Means method

Figure 4.9 shows that fuel consumption will be divided into several clusters (more precisely 5) regarding ship speed. This cluster will make it easier for users to process data which will later be used in other methods such as regression or more complex methods.

4.3.3.2 Wind Power

In this method, the wind strength is used as a reference which will later be classified with specific parameters of fuel consumption that have been obtained.

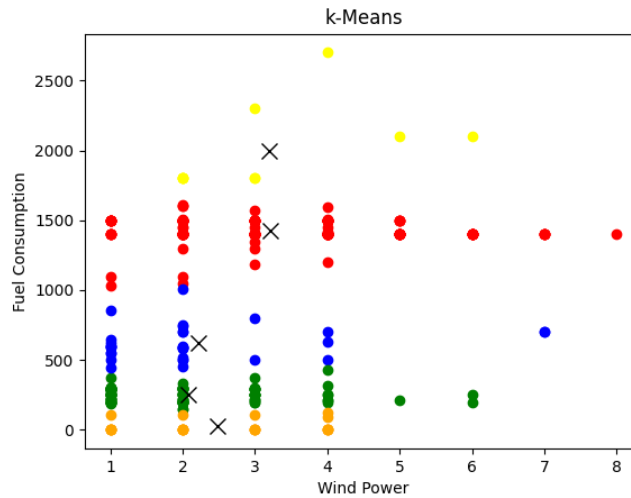


Figure 4.2 Wind Power variable output from k Means method

Figure 4.10 shows that fuel consumption will be divided into several clusters (more precisely 5) regarding wind strength. This cluster will make it easier for users to process data which will later be used in other methods such as regression or more complex methods.

4.3.3.3 Sea Condition

In this method, the sea condition is used as a reference which will later be classified with specific parameters of fuel consumption that have been obtained.

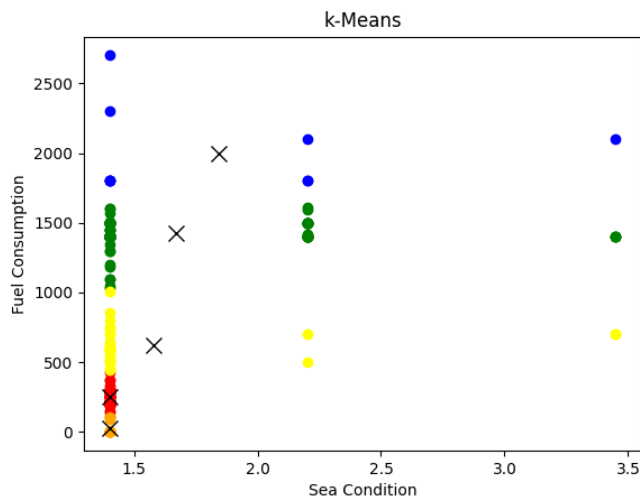


Figure 4.3 Sea Condition variable output from k Means method

Figure 4.11 shows that fuel consumption will be divided into several clusters (more precisely 5) regarding sea conditions. This cluster will make it easier for users to process data which will later be used in other methods such as regression or more complex methods.

4.3.3.4 Engine RPM

In this method, the engine's rpm is used as a reference which will later be classified with specific parameters of fuel consumption that have been obtained.

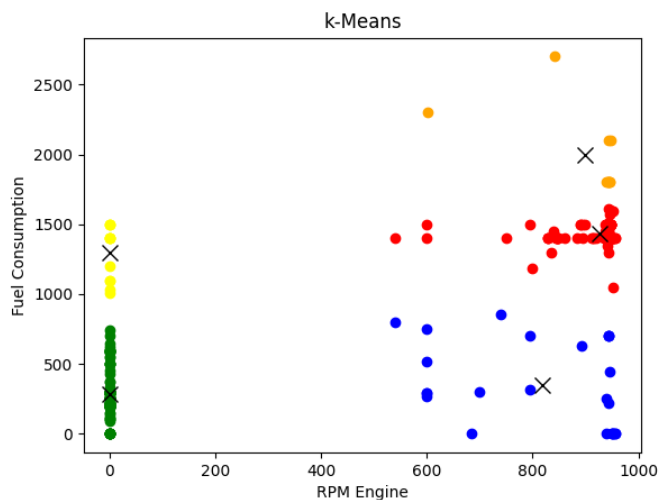


Figure 4.4 Engine RPM variable output from k Means method

Figure 4.12 shows that fuel consumption will be divided into several clusters (more precisely 5) regarding engine rpm. This cluster will make it easier for users to process data which will later be used in other methods such as regression or more complex methods.

4.4 Regression Model

This method will focus on the estimation of fuel consumption on the ship, using four variables, using three regression methods. The resulting output is a fuel estimate, MAE (Mean Absolute Error), the limit of the estimate, and R Squared, which is the accuracy of the method used. The results can be summarized in the tables below from several regressions.

Table 4.4 The results of the calculation of the Linear Regression method

No	Variables	MAE ($\pm$)	R^2 (%)
1	4 Variables (Ship Speed, Wind Power, Sea Condition, Engine RPM)	151.99	29.4

Table 4.5 The results of the calculation of the Polynomial Regression method

No	Variables	MAE ($\pm$)	R^2 (%)
1	4 Variables (Ship Speed, Wind Power, Sea Condition, Engine RPM)	13.66	98.99

Table 4.6 The results of the calculation of the Support Vector Regression method

No	Variables	MAE ($\pm$)	R^2 (%)
1	4 Variables (Ship Speed, Wind Power, Sea Condition, Engine RPM)	107.26	-

The tables above show the calculation results of each regression method. The results obtained are Mean Absolute Error (MAE) and R-Squared. Where MAE is the average error of the calculation and r-squared is the accuracy of the model that has been used. Regression in this study used three methods: Multilinear, Polynomial, and Support Vector. Each method has advantages and disadvantages in the calculation.

From the calculation results above, it can be determined that polynomial regression has the highest accuracy value compared to other models. For results with four variables as a calculation parameter, an accuracy of 98.99% is obtained. This result is quite good because it is close to perfect results for three variables as parameters,

including ship speed, wind power, and engine rpm. The accuracy value obtained is 98.22%. Two variables as parameters, including ship speed and wind power. The accuracy value obtained is 93.67%. One variable as a parameter that significantly affects the calculation of fuel consumption is ship speed, with an accuracy value of 91.35%.

The results of the linear regression calculations are very different from the results of polynomial regression, where the accuracy obtained does not exceed 40%. This indicates that the relationship between the four variables and fuel consumption does not move linearly but is polynomial. And this is right with the theory that the speed of the ship moves polynomial to fuel consumption, where these parameters obtain the highest accuracy compared to other parameters.

4.4.1 Linear Regression

This method uses all the variables that become the reference in determining a ship's fuel consumption. By using many variables, you will get various outputs.

4.4.1.1 4 Variables (Ship Speed, Wind Power, Sea Condition, Engine RPM)

Four variables were used to calculate the Multilinear regression method, namely ship speed, wind strength, sea conditions, and engine rpm. In the programming, 75% of all data is used as training data which is expected to produce predictions with reasonable accuracy, and 25% of all data is used as testing data to check the results of previous training.

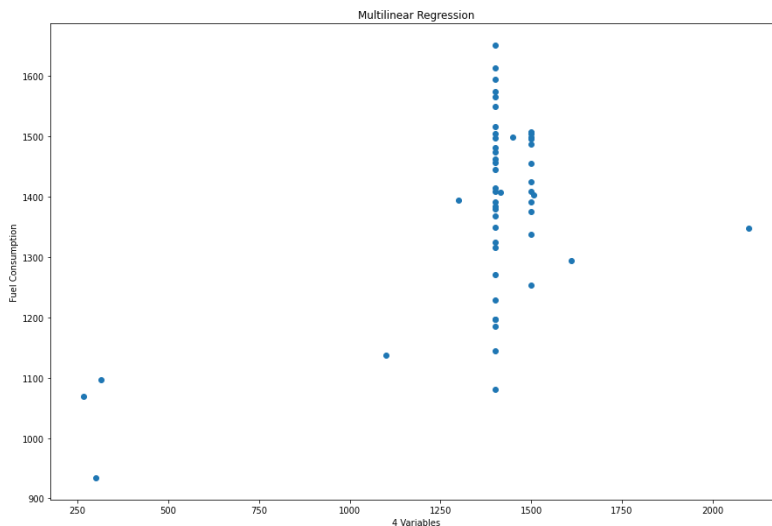


Figure 4.5 Multilinear regression scatter plot of all variables as features

Table 4.7 Describe all variables as features

index	Fuel Actual	Fuel Predict	Residual	Difference (%)
count	53	53	53	53
mean	1374.83	1382.48	-7.65	21.70
std	291.25	153.60	244.55	58.14
min	268	933.97	-801.12	0.04
25%	1400	1316.02	-80.90	2.94
50%	1400	1407.22	3.97	6.81
75%	1500	1497.37	102.56	13.91
max	2100	1651.47	751.55	298.93

Figure 4.13 shows a data plot of the four variables used. This plot is used to train using a linear regression method. The data training and data testing results have resulted in a brief explanation of the calculation of these four variables, which can be seen in table 4.7. The difference between the actual data and the predicted data yields the lowest number in 0.04% and the highest number in 298.93%, for the average difference is 21.7%.

From the programming carried out, a formula related to the estimation of the required fuel is obtained, it can be seen in the formula below. And attachment 5 explains in detail the algorithmic flow of the linear regression method.

$$y = 982.73 + 60.92(x_1) - 40.159(x_2) - 69.787(x_3) - 0.114(x_4)$$

Notes:

y = Fuel Consumption

x1 = Ship Speed Variable

x2 = Wind Power Variable

x3 = Sea Condition Variable

x4 = Engine RPM Variable

4.4.2 Polynomial Regression

The polynomial method is not much different from the linear method. The distinguishing feature is that in this method, a quadratic curve or more is used to follow the shape of the scattered data. Four variables are used in this method which will be used for the final prediction of ship fuel.

4.4.2.1 4 Variables (Ship Speed, Wind Power, Sea Condition, Engine RPM)

Four variables were used to calculate the Polynomial regression method: ship speed, wind strength, sea conditions, and engine rpm. In the programming, 80% of all data is used as training data which is expected to produce predictions with reasonable accuracy, and 20% of all data is used as testing data to check the results of previous training.

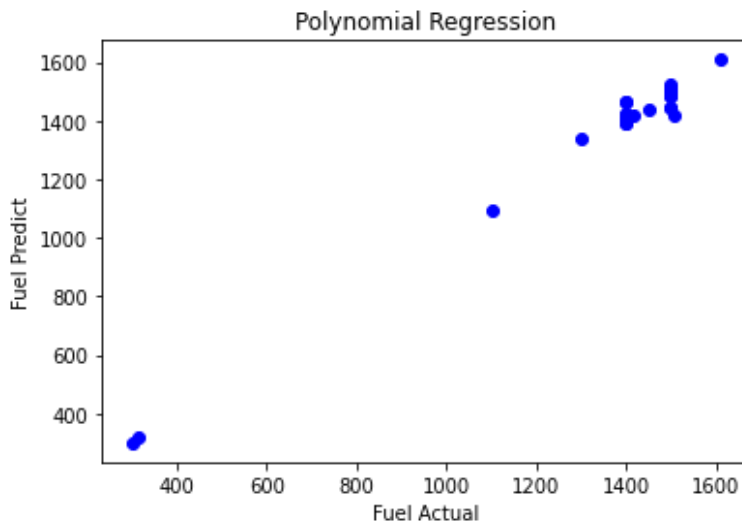


Figure 4.6 Polynomial regression scatter plot of all variables as features

Table 4.8 Describe of all variables as features

index	Fuel Actual	Fuel Predict	Residual	Difference (%)
count	43	43	43	43
mean	1376.70	1376.70	2.87E-09	0.95
std	250.70	249.47	25.15	1.44
min	300	300.53	-62.20	0.03
25%	1400	1399.22	-3.06	0.11
50%	1400	1412.43	0.39	0.25
75%	1500	1472.43	3.66	0.94
max	1610	1608.44	88.87	5.90

Figure 4.28 shows a data plot of the four variables used. This plot is used to train using a polynomial regression method. The data training and data testing results have resulted in a brief explanation of the calculation of these four variables, which can be seen in table 4.8. The difference between the actual data and the predicted data yields the lowest number in 0.03% and the highest number in 5.9%, for the average difference is in the 0.95% number.

A slope and intercept that can be used as a formula related to the estimation of the required fuel are obtained from the programming carried out. It can be seen in the information below.

Slope : 3.41749721e+03, -6.37522552e+02, 4.91421203e+01, 3.68706895e+01, 3.13527072e+03, -3.20225539e+04, -3.84533062e+03, -4.43723498e+02, 1.04819975e+03, 2.79827903e+03, 1.69977767e+02, 7.67807375e+00, -1.66506069e-02, 2.34573089e+03, -1.07330755e+01, 4.73741816e+02, -1.92510704e+03, -2.40191575e+04, 6.59900492e+01, 2.14532680e+02, -2.88509392e+03, 4.81311538e+01, -3.33154983e+02, 6.27073438e+02, -2.50766353e+00, 1.46254250e+03, 2.09843263e+03, -2.14468884e+01, 1.27388021e+02, -1.93926463e+01, -1.76222971e-01, -9.06669574e-03, 1.75824396e+03, -9.63393943e+00, 1.93192425e-02, -2.36390438e+00, 1.81907877e+01, 2.84403067e+02, -6.51450144e-01, 1.17929172e+02, -1.44855745e+03, 2.13137573e+00, -1.80189902e+04, 6.13179550e+01, -5.15149848e-02, -1.82040320e+02, 1.59407167e+02, -1.75224691e+00, -2.16518359e+03, 4.64994023e+01, -6.55280798e-02, -2.50558142e+02, 2.47456343e+02, -1.24387457e+00, 1.78163806e-03, -1.52948827e+02, 1.09444833e+03, 1.35546640e+00, 1.57344110e+03, -1.42772434e+01, 2.20463863e-02, 9.54247092e+01, -4.97583852e+01, -1.12161785e-01, 2.73264606e-04, -1.36718565e-02, 1.31720778e+03, -8.55444116e+00, 1.80056106e-02, -1.52437929e-05

Intercept : -1276667.61206786

The slope (steepness) and y-intercept (the line crosses the vertical y-axis) are simple ways to make a formula. The author does not make a formula from the slope-intercept obtained because of the formula's complex form, which will confuse the author and others. And attachment 6 explains in detail the algorithmic flow of the linear regression method.

4.4.3 Support Vector Regression

This method is different from the linear and polynomial methods. In this method, the scattered data can be the boundary of a curve whose purpose is to support getting more accurate results. In this method, four variables will be used one by one in predicting the final value. Attachment 7 explains in detail the algorithmic flow of the linear regression method.

4.4.3.1 4 Variables (Ship Speed, Wind Power, Sea Condition, Engine RPM)

Four variables are used to calculate the support vector regression method. This method does not separate training and testing data but makes all data be testing data. There is quite a lot for the data, and the location is random, so it is a bit more difficult to use this method.

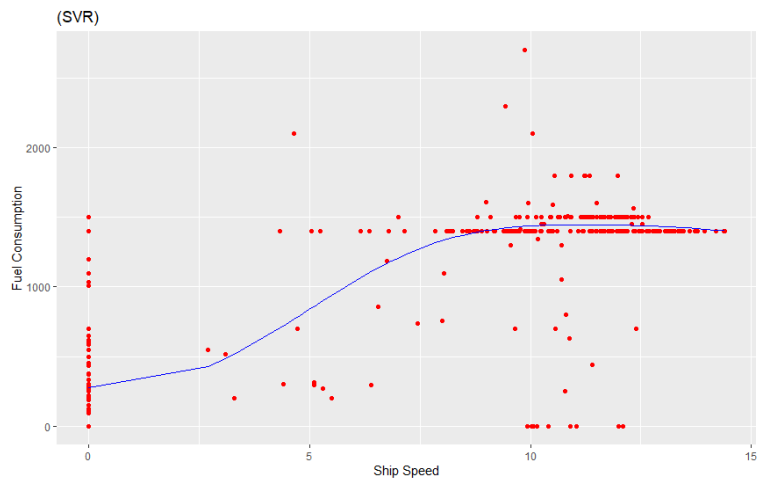


Figure 4.7 Support vector regression scatter plot of ship speed as a feature

Figure 4.43 shows a data plot of the variable speed of the ship used. This plot is used to train using the support vector regression method. In this plot, a radial line is used to adjust to the shape of the scattered data.

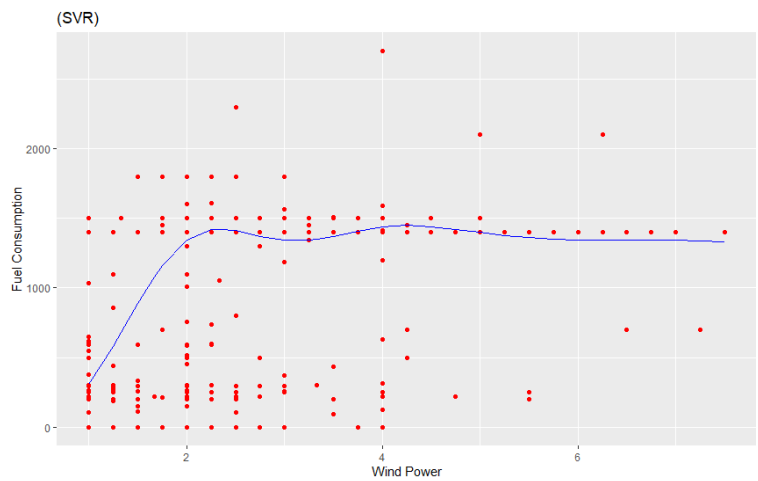


Figure 4.8 Support vector regression scatter plot of wind power as a feature

Figure 4.44 shows a data plot of the wind power variable used. This plot is used to train using the support vector regression method. In this plot, a radial line is used to adjust to the shape of the scattered data.

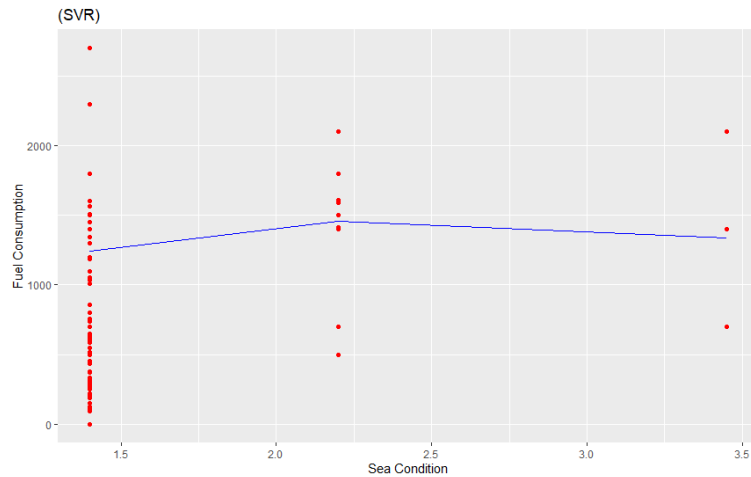


Figure 4.9 Support vector regression scatter plot of sea condition as a feature

Figure 4.45 shows a data plot of the sea condition variable used. This plot is used to train using the support vector regression method. In this plot, a radial line is used to adjust to the shape of the scattered data.



Figure 4.10 Support vector regression scatter plot of engine rpm as a feature

Figure 4.46 shows a data plot of the engine rpm variable used. This plot is used to train using the support vector regression method. In this plot, a radial line is used which will adjust to the shape of the scattered data.

The algorithm of the SVR method is different from the linear or polynomial regression method. In this method, all data is used in the calculation. Therefore, the hypothesis regarding accuracy and error is better. However, after the calculations, it was found that the error obtained was still more incredible than the polynomial

regression and smaller than the linear regression, which in this method resulted in an error of 107.26 L.

4.5 Website Prototype

In this study, the Indonesian Navy requested a prototype website or application to facilitate the calculation of fuel estimates on the KRI Bima Suci ship. Therefore, we need a front-end with the user interface and user experience. In this study, the platform used in the frontend process is Figma. As for the backend, which is programming to run well, the platforms used are various programming languages such as JavaScript and Python.

4.5.1 User Flow

User flow is a sequence carried out by a user when running a website or application. You could say this user flow is a sitemap which is the structure of the website or application and determines a temporary information structure to make it easier to create wireframes. You can see the user flow of the website prototype for this study below.

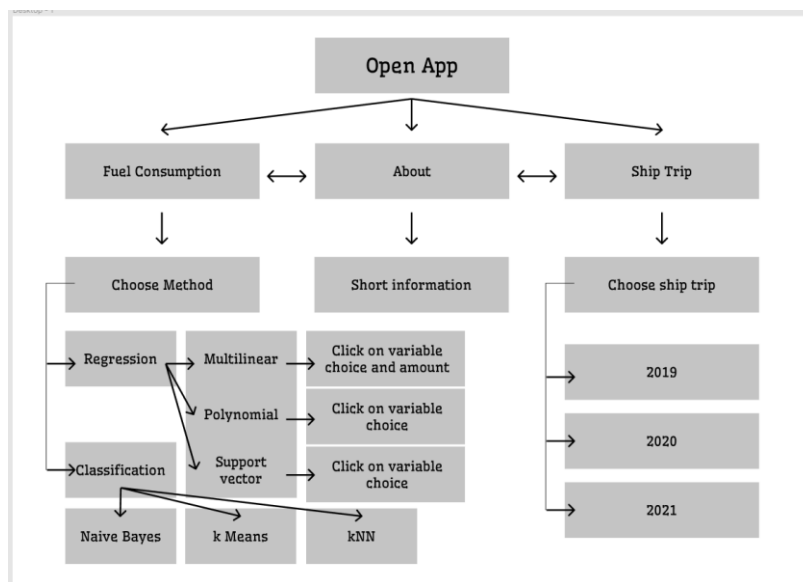


Figure 4.11 User flow of website prototype

4.5.2 Wireframe

A wireframe is a simple sketch made from the basis of user flow. This aims to make it easy for designers to determine what information will appear on a website or

application and structure content and layouts made in such a way. From this wireframe, a more complex mock-up will be created.



Figure 4.12 Wireframe of a website prototype

4.5.3 UI/UX Design

In designing the user interface and user experience using the Figma website as the platform, designers are needed in this process because they must impress users with good features and attractive designs.

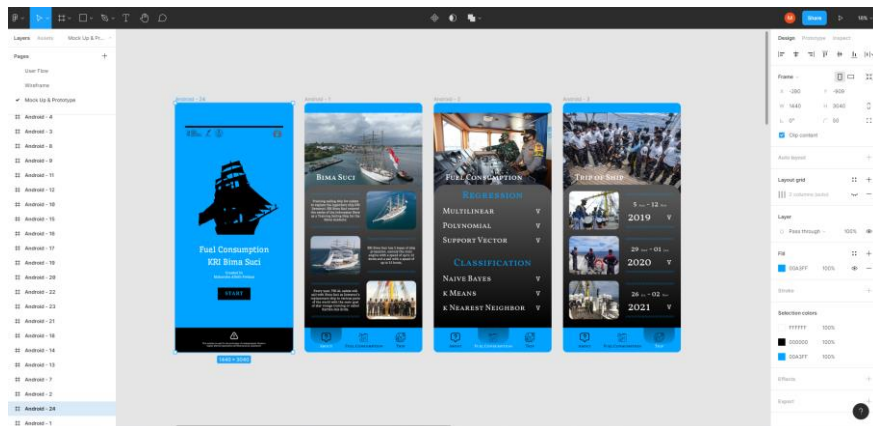


Figure 4.13 UI/UX design by Figma of a website prototype

The mock-up is a continuation of the wireframe, given more detailed color, content, and design. You could say this mock-up is the finished product of a website or application that the user will use. A product simulation and animation or transition on the screen from this mock-up have also been given.

4.5.4 Prototype

The prototype of this website serves to facilitate users related to information about KRI Bima Suci. The three menus in this prototype include brief details on the ship, the ship's annual trip schedule, and the most important thing is the estimation of fuel on the ship, which is the main menu in this study. The backend has not been combined with the frontend in this website because the ship's data is confidential. Therefore, the Indonesian Navy asks not to be combined first. This website prototype is specifically for smartphone users because it is easy to carry anywhere and does not require a large space. The desktop view will be created if the Indonesian Navy requests it.

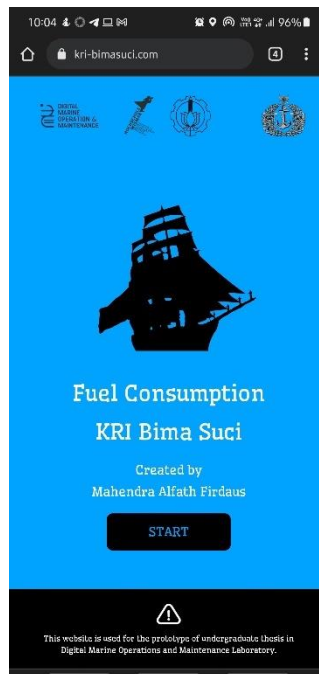


Figure 4.14 Homepage website prototype



Figure 4.15 About menu of a website prototype



Figure 4.16 Fuel consumption menu of a website prototype

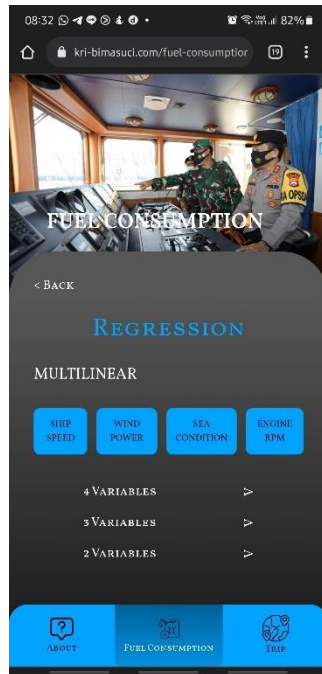


Figure 4.17 Regression menu of a website prototype



Figure 4.18 Classification menu of a website prototype



Figure 4.19 Trip Menu of a website prototype

4.6 Discussion

Estimating ship fuel requires many parameters that will affect it, starting from macro to micro factors. Only four parameters are used in this study, namely ship speed, wind power, sea condition, and engine rpm. The calculations carried out are considered very good, especially in the polynomial regression method. Still, in reality, the resulting formula is only a statistical formula that does not use all the variables related to fuel consumption. Therefore, much remains to be involved in these calculations. The difficulty of obtaining data on related variables is a reason for the less than optimal use of the formula in actual reality. The author hopes to be directly involved in monitoring fuel consumption so that the variables involved can be obtained optimally.

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CHAPTER 5

CONCLUSION

5.1 Overview of the Research

In the overview of this research, it is explained that the background of this research is the problem in estimating fuel consumption on the KRI Bima Suci due to manual calculations. Therefore, this research focuses on helping the KRI Bima Suci get a fuel estimate with good accuracy. By using machine learning methods and the existence of a website, it is hoped that it will make it easier for KRI Bima Suci to deal with fuel problems.

5.2 Conclusion

In this study, the regression model uses three methods, including linear regression, polynomial regression, and support vector regression, to estimate fuel for the KRI Bima Suci engine. For the classification model, there are three, including Naive Bayes, k Nearest Neighbor, and k Means. For Naive Bayes and k Nearest Neighbor can be done for prediction, but in this study, it is only used for data classification, while k Means is purely for clustering. The conclusions from this research are:

1. This study found that many variables affect the ship's fuel consumption. The variables used in this study include 4 data, both internal and external variables. Ship speed is the most crucial variable among the four variables used because ship speed will affect the performance of the ship's engine. In reality, many other variables affect fuel consumption, such as draft, engine load, displacement, and others. However, it isn't easy to obtain these data in field conditions.
2. Machine learning modeling consists of three classification methods and three regression methods. The programming carried out in this study uses the python language as a language platform in the machine learning process. For a prototype website using the Figma platform and JavaScript.
3. Machine learning methods that are very effective in classification and regression are the K Nearest Neighbor method for classification and the Polynomial Regression method for regression. From the calculations, the accuracy of the K Nearest Neighbor method is 84.95%, while the Naive Bayes method is only 79.57%. The polynomial regression method is 98.99% for the regression calculation

accuracy with an average error of only 13.66 L. As for the Linear Regression method, the accuracy obtained is 29.4%, with an average error of 151.99 L. And the Support Vector regression method produces an average error of 107.26 L.

5.3 Recommendation

Based on this research, some recommendation is given by the author to support future research, as follow:

1. For the calculation results to produce good accuracy and minimize errors, it would be better to increase the data variables used, or the amount of existing data was reproduced.
2. The regression method is added to other types so that the user can get more choices in calculating the estimated fuel consumption, and in using the classification method, different types are added to make it easier for the user.
3. It is recommended to use an Android, iOS, Windows, or Mac application instead of a website because the user will be more flexible and safer.
4. In making it easier for users to use the existing website, it is necessary to develop different entry levels such as the operational and engine divisions, ship commanders, and the public. The operating and machine divisions and ship commanders will require a specific login and different data processing features.

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Appendix

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ATTACHMENT 1
Ship logbook and noon report example

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PENGISIAN DAN PEMAKAIAN LOGGA

HARI ANGGAL	JAM	BAHAN BAKAR					AIR TAWAR					OLI				KETERANGAN			
		11	12	32	33	JUMLAH	%	5	6	13	14	JUMLAH	%	Gardun 40	%		Shell 30	%	Remula 15/40
SENIN 02-08-2021	04.00					01.324	65.70					75.606	60.06						
	08.00					88.924	64.6					75.080	60.4						
	12.00					88.524	63.6					75.080	60.4						
	16.00					89.124	62.69					71.000	57.25						
	20.00					85.724	61.6					67.000	54.0						
SELASA 03-08-2021	04.00					84.324	60.6					76.400	56.4						
	08.00					82.924	59.5					76.400	56.4						
	12.00					82.524	58.6					70.000	56.4						
	16.00					80.124	57.6					71.660	57.7						
	20.00					78.724	56.6					71.000	57.25						
RABU 04-08-2021	04.00					77.324	55.6					72.800	58.7						
	08.00					75.924	54.6					73.000	58.8						
	12.00					74.524	53.61					70.000	54.45						
	16.00					73.124	52.6					72.500	58.46						
	20.00					71.724	51.6					72.100	58.1						
KAMIS 05-08-2021	04.00					70.324	50.69					68.000	54.81						
	08.00					68.924	49.58					70.000	56.45						
	12.00					67.524	48.5					69.000	55.9						
	16.00					66.124	47.5					67.700	54.59						
	20.00					64.724	46.5					67.000	54.0						
JUMAT 06-08-2021	04.00					63.324	45.5					66.400	53.5						
	08.00					61.924	44.5					65.000	52.7						
	12.00					60.524	43.5					63.600	51.7						
	16.00					59.124	42.5					62.200	50.7						
	20.00					57.724	41.5					60.800	49.7						

ATTACHMENT 2
Algorithm of K Nearest Neighbor



```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Import some libraries to perform certain functions, where NumPy to help with numerical, matplotlib for data visualization, and pandas to help with datasets.



```
dataset = pd.read_excel('Data Angka.xlsx')
data = dataset.replace(np.NaN, dataset.mean())
X = data.drop(['Done'],axis=1).values
y = data['Done'].values
```

Pandas library will import the dataset. The average of certain variables fills in the blank data. Fuel consumption functions as Y as the independent variable, and the other four variables function as X as the dependent variable.



```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
```

Separate training data and testing data to perform calculations. Of all the data used, 75% of the training data and 25% of the testing data are used.



```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Import the standard scaler function, which is helpful for generalizing numbers later aligned in plot visualizations to make the training data easier to use.

```
from sklearn.neighbors import KNeighborsClassifier
classifier = KNeighborsClassifier(n_neighbors = 5, metric = 'minkowski', p = 2)
classifier.fit(X_train, y_train)
```

Import the K Nearest Neighbor function to calculate the generalized training data. By using the five closest data as a reference.

```
print(classifier.predict(sc.transform([[10, 4, 0.75, 800, 1000]])))
y_pred = classifier.predict(X_test)
print(np.concatenate((y_pred.reshape(len(y_pred),1), y_test.reshape(len(y_test),1)),1))
```

Predict in detail by inputting several variables or assessing the overall calculation of the testing data.

```
from sklearn.metrics import confusion_matrix, accuracy_score
cm = confusion_matrix(y_test, y_pred)
print(cm)
accuracy_score(y_test, y_pred)
```

Generate a confusion matrix to assess the accuracy of the classification calculation. Data testing is a reference for the confusion matrix as many as 93 data.

```
from matplotlib.colors import ListedColormap
X_set, y_set = sc.inverse_transform(X_test), y_test
X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 10, stop = X_set[:, 0].max() + 10, step = 1),
                     np.arange(start = X_set[:, 1].min() - 1000, stop = X_set[:, 1].max() + 1000, step = 1))
plt.contourf(X1, X2, classifier.predict(sc.transform(np.array([X1.ravel(), X2.ravel()]).T)).reshape(X1.shape),
             alpha = 0.75, cmap = ListedColormap(('red', 'green')))
plt.xlim(X1.min(), X1.max())
plt.ylim(X2.min(), X2.max())
for i, j in enumerate(np.unique(y_set)):
    plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1], c = ListedColormap(('red', 'green'))(i), label = j)
plt.title('K-NN')
plt.xlabel('Engine RPM')
plt.ylabel('Fuel Consumption')
plt.legend()
plt.show()
```

Visualize numerical data where data testing will act as the X-axis and Y-axis to create a graph plot to make it easier for the user.

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ATTACHMENT 3
Algorithm of Naïve Bayes



```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Import some libraries to perform certain functions, where NumPy to help with numerical, matplotlib for data visualization, and pandas to help with datasets.



```
dataset = pd.read_excel('Data Angka.xlsx')
data = dataset.replace(np.NaN, dataset.mean())
X = data.drop(['Done'],axis=1).values
y = data['Done'].values
```

Pandas library will import the dataset. The average of certain variables fills in the blank data. Fuel consumption functions as Y as the independent variable, and the other four variables function as X as the dependent variable.



```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
```

Separate training data and testing data to perform calculations. Of all the data used, 75% of the training data and 25% of the testing data are used.



```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Import the standard scaler function, which is helpful for generalizing numbers later aligned in plot visualizations to make the training data easier to use.

```
from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)
```

Import the Naïve Bayes function to calculate the generalized training data. By using the five closest data as a reference.

```
print(classifier.predict(sc.transform([[10, 4, 0.75, 800, 1000]])))
y_pred = classifier.predict(X_test)
print(np.concatenate((y_pred.reshape(len(y_pred),1), y_test.reshape(len(y_test),1)),1))
```

Predict in detail by inputting several variables or assessing the overall calculation of the testing data.

```
from sklearn.metrics import confusion_matrix, accuracy_score
cm = confusion_matrix(y_test, y_pred)
print(cm)
accuracy_score(y_test, y_pred)
```

Generate a confusion matrix to assess the accuracy of the classification calculation.
Data testing is a reference for the confusion matrix as many as 93 data.

```
from matplotlib.colors import ListedColormap
X_set, y_set = sc.inverse_transform(X_test), y_test
X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 10, stop = X_set[:, 0].max() + 10, step = 1),
                     np.arange(start = X_set[:, 1].min() - 1000, stop = X_set[:, 1].max() + 1000, step = 1))
plt.contourf(X1, X2, classifier.predict(sc.transform(np.array([X1.ravel(), X2.ravel()]).T)).reshape(X1.shape),
             alpha = 0.75, cmap = ListedColormap(('red', 'green')))
plt.xlim(X1.min(), X1.max())
plt.ylim(X2.min(), X2.max())
for i, j in enumerate(np.unique(y_set)):
    plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1], c = ListedColormap(('red', 'green'))(i), label = j)
plt.title('K-NN')
plt.xlabel('Engine RPM')
plt.ylabel('Fuel Consumption')
plt.legend()
plt.show()
```

Visualize numerical data where data testing will act as the X-axis and Y-axis to create a graph plot to make it easier for the user.

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ATTACHMENT 4
Algorithm of K Means

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Import some libraries to perform certain functions, where NumPy to help with numerical, matplotlib for data visualization, and pandas to help with datasets.

```
dataset = pd.read_excel('Data Angka.xlsx')
data = dataset.replace(np.NaN, dataset.mean())
X = data.drop(['Done'],axis=1).values
y = data['Done'].values
```

Pandas library will import the dataset. The average of certain variables fills in the blank data. Fuel consumption functions as Y as the independent variable, and the other four variables function as X as the dependent variable.

```
from sklearn.cluster import KMeans
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters = i, init = 'k-means++', random_state = 42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)
plt.plot(range(1, 11), wcss)
plt.title('The Elbow Method')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()
```

Importing the K Means function to perform a clustering simulation carried out on existing data.

```
kmeans = KMeans(n_clusters = 5, init = 'k-means++', random_state = 42)
y_kmeans = kmeans.fit_predict(X)
```

Entering existing data into the k means function.

```
plt.scatter(X[y_kmeans == 0, 0], X[y_kmeans == 0, 1], s = 100, c = 'red', label = 'Cluster 1')
plt.scatter(X[y_kmeans == 1, 0], X[y_kmeans == 1, 1], s = 100, c = 'blue', label = 'Cluster 2')
plt.scatter(X[y_kmeans == 2, 0], X[y_kmeans == 2, 1], s = 100, c = 'green', label = 'Cluster 3')
plt.scatter(X[y_kmeans == 3, 0], X[y_kmeans == 3, 1], s = 100, c = 'cyan', label = 'Cluster 4')
plt.scatter(X[y_kmeans == 4, 0], X[y_kmeans == 4, 1], s = 100, c = 'magenta', label = 'Cluster 5')
plt.scatter(kmeans.cluster_centers[:, 0], kmeans.cluster_centers[:, 1], s = 300, c = 'yellow', label =
'Centroids')
plt.title('Clusters of Fuel Consumption')
plt.xlabel('Fuel Consumption')
plt.ylabel('4 variables')
plt.legend()
plt.show()
```

Plotting data for graphic visualization where the fuel consumption variable is the Y-axis, and four other variables are the X-axis.

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ATTACHMENT 5
Algorithm of Linear Regression



```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Import some libraries to perform certain functions, where NumPy to help with numerical, matplotlib for data visualization, and pandas to help with datasets.



```
data_df = pd.read_excel('Data Angka.xlsx', sheet_name='Sheet3')
data_df = data_df.replace(np.NaN, data_df.mean())
data_df.head()
```

Pandas library will import the dataset. The average of certain variables fills in the blank data.



```
x = data_df.drop(['Fuel Consumption'],axis=1).values
y = data_df['Fuel Consumption'].values
```

Fuel consumption functions as Y as the independent variable, and the other four variables function as X as the dependent variable.



```
from sklearn.model_selection import train_test_split
train_x, test_x, train_y, test_y=train_test_split(x, y, test_size=0.25, random_state=0)
```

Separate training data and testing data to perform calculations. Of all the data used, 75% of the training data and 25% of the testing data are used.

```
from sklearn.linear_model import LinearRegression
lin = LinearRegression()
lin.fit(train_x, train_y)
```

Import linear regression functions to perform calculations on training data.

```
from sklearn.metrics import r2_score, mean_absolute_error
r2_score(test_y, y_pred)
mean_absolute_error(test_y, y_pred)
ml.coef_
ml.intercept_
```

Import the sklearn function to determine the accuracy and error in calculations and find out the slope and intercept used to get a formula.

```
import matplotlib.pyplot as plt
plt.figure(figsize=(15,10))
plt.scatter(test_y, y_pred)
plt.title('Multilinear Regression')
plt.xlabel('Fuel Actual')
plt.ylabel('Fuel Predict')
```

Plotting data testing for graphic visualization where the fuel consumption variable is the Y-axis, and four other variables are the X-axis.

```
pred_y_df = pd.DataFrame({'Fuel Actual':test_y, 'Fuel Predict':y_pred, 'Residual': test_y-y_pred, 'Diference (%)':
np.absolute(test_y-y_pred)/test_y*100})
pred_y_df
pred_y_df.describe()
```

See the overall difference and a brief overview between fuel prediction and actual fuel.

index	Fuel Actual	Fuel Predict	Residual	Difference (%)
0	1500	1391.72	108.28	7.22
1	1400	1594.76	-194.76	13.91
2	1416	1407.22	8.78	0.62
3	300	933.97	-633.97	211.32
4	1400	1185.60	214.40	15.31
5	1400	1574.76	-174.76	12.48
6	1450	1499.23	-49.23	3.40
7	1610	1294.53	315.47	19.59
8	1400	1349.91	50.09	3.58
9	1400	1549.51	-149.51	10.68
10	1400	1463.18	-63.18	4.51
11	1400	1565.90	-165.90	11.85
12	1100	1137.65	-37.65	3.42
13	1500	1505.26	-5.26	0.35
14	1400	1271.62	128.38	9.17
15	1500	1253.95	246.05	16.40
16	1400	1497.37	-97.37	6.95
17	1400	1413.98	-13.98	1.00
18	1506	1403.44	102.56	6.81
19	1500	1455.97	44.03	2.94
20	1400	1456.82	-56.82	4.06
21	1400	1474.86	-74.86	5.35
22	1500	1425.51	74.49	4.97
23	1400	1504.86	-104.86	7.49
24	316	1097.10	-781.10	247.18
25	1500	1506.71	-6.71	0.45
26	1400	1316.02	83.98	6.00
27	1400	1080.69	319.31	22.81
28	1400	1480.90	-80.90	5.78
29	1400	1391.20	8.80	0.63
30	1500	1409.59	90.41	6.03
31	1400	1144.36	255.64	18.26
32	1300	1393.73	-93.73	7.21
33	1400	1383.77	16.23	1.16
34	1500	1375.81	124.19	8.28
35	1400	1229.19	170.81	12.20

36	1500	1337.05	162.95	10.86
37	1500	1499.39	0.61	0.04
38	1500	1508.03	-8.03	0.54
39	1400	1445.19	-45.19	3.23
40	1400	1368.54	31.46	2.25
41	1500	1496.03	3.97	0.26
42	1400	1196.72	203.28	14.52
43	1400	1409.31	-9.31	0.67
44	1500	1487.88	12.12	0.81
45	1400	1197.45	202.55	14.47
46	1400	1380.37	19.63	1.40
47	268	1069.12	-801.12	298.93
48	1400	1516.55	-116.55	8.33
49	1400	1614.02	-214.02	15.29
50	2100	1348.45	751.55	35.79
51	1400	1324.98	75.02	5.36
52	1400	1651.47	-251.47	17.96

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ATTACHMENT 6
Algorithm of Polynomial Regression

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Import some libraries to perform certain functions, where NumPy to help with numerical, matplotlib for data visualization, and pandas to help with datasets.

```
data_df = pd.read_excel('Data Angka.xlsx', sheet_name='Sheet3')
data_df = data_df.replace(np.NaN, data_df.mean())
data_df.head()
```

Pandas library will import the dataset. The average of certain variables fills in the blank data.

```
x = data_df.drop(['Fuel Consumption'],axis=1).values
y = data_df['Fuel Consumption'].values
```

Fuel consumption functions as Y as the independent variable, and the other four variables function as X as the dependent variable.

```
from sklearn.model_selection import train_test_split
train_x, test_x, train_y, test_y=train_test_split(x, y, test_size=0.25, random_state=0)
```

Separate training data and testing data to perform calculations. Of all the data used, 75% of the training data and 25% of the testing data are used.

```
from sklearn.linear_model import LinearRegression
lin = LinearRegression()
lin.fit(train_x,train_y)
```

Import linear regression functions to perform calculations on training data

```
from sklearn.preprocessing import PolynomialFeatures
pol = PolynomialFeatures(degree = 4)
x_pol = pol.fit_transform(test_x)
lin2 = LinearRegression()
lin2.fit(x_pol, test_y)
```

The previously obtained linear function is developed into a polynomial function with fixed training data as a reference.

```
from sklearn.metrics import r2_score, mean_absolute_error
r2_score(test_y, y_pred)
mean_absolute_error(test_y, y_pred)
ml.coef_
ml.intercept_
```

Import the sklearn function to find out the accuracy and error in calculations, and find out the slope and intercept used to get a formula

```
import matplotlib.pyplot as plt
plt.figure(figsize=(15,10))
plt.scatter(test_y, y_pred)
plt.title('Multilinear Regression')
plt.xlabel('Fuel Actual')
plt.ylabel('Fuel Predict')
```

Plotting data testing for graphic visualization where the fuel consumption variable is the Y-axis and the other four variables are the X-axis.

```
pred_y_df = pd.DataFrame({'Fuel Actual':test_y, 'Fuel Predict':y_pred, 'Residual': test_y-y_pred, 'Diference (%)':
np.absolute(test_y-y_pred)/test_y*100})
pred_y_df
pred_y_df.describe()
```

See the overall difference as well as a brief overview between fuel prediction and actual fuel.

index	Fuel Actual	Fuel Predict	Residual	Difference (%)
0	1500	1501.67	-1.67	0.11
1	1400	1402.46	-2.46	0.18
2	1416	1415.39	0.61	0.04
3	300	300.53	-0.53	0.18
4	1400	1395.43	4.57	0.33
5	1400	1393.06	6.94	0.50
6	1450	1438.90	11.10	0.77
7	1610	1608.44	1.56	0.10
8	1400	1391.58	8.42	0.60
9	1400	1400.49	-0.49	0.04
10	1400	1462.20	-62.20	4.44
11	1400	1412.43	-12.43	0.89
12	1100	1097.26	2.74	0.25
13	1500	1482.66	17.34	1.16
14	1400	1403.35	-3.35	0.24
15	1500	1502.39	-2.39	0.16
16	1400	1461.69	-61.69	4.41
17	1400	1399.61	0.39	0.03
18	1506	1417.13	88.87	5.90
19	1500	1442.99	57.01	3.80
20	1400	1399.17	0.83	0.06
21	1400	1427.91	-27.91	1.99
22	1500	1527.25	-27.25	1.82
23	1400	1406.76	-6.76	0.48
24	316	316.65	-0.65	0.20
25	1500	1493.47	6.53	0.44
26	1400	1415.58	-15.58	1.11
27	1400	1399.27	0.73	0.05
28	1400	1413.82	-13.82	0.99
29	1400	1399.15	0.85	0.06
30	1500	1512.81	-12.81	0.85
31	1400	1402.76	-2.76	0.20
32	1300	1335.77	-35.77	2.75
33	1400	1400.53	-0.53	0.04
34	1500	1497.66	2.34	0.16
35	1400	1401.83	-1.83	0.13

index	Fuel Actual	Fuel Predict	Residual	Difference (%)
36	1500	1498.67	1.33	0.09
37	1500	1487.31	12.69	0.85
38	1500	1443.20	56.80	3.79
39	1400	1400.75	-0.75	0.05
40	1400	1398.49	1.51	0.11
41	1500	1491.24	8.76	0.58
42	1400	1398.31	1.69	0.12

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ATTACHMENT 7
Algorithm of Support Vector Regression



```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Import some libraries to perform certain functions, where NumPy to help with numerical, matplotlib for data visualization, and pandas to help with datasets.



```
data_df = pd.read_excel('Data Angka.xlsx', sheet_name='Sheet3')
data_df = data_df.replace(np.NaN, data_df.mean())
data_df.head()
```

Pandas library will import the dataset. The average of certain variables fills in the blank data.



```
x = data_df.drop(['Fuel Consumption'],axis=1).values
y = data_df['Fuel Consumption'].values
```

Fuel consumption functions as Y as the independent variable, and the other four variables function as X as the dependent variable.



```
from sklearn.preprocessing import StandardScaler
sc_X = StandardScaler()
sc_y = StandardScaler()
X = sc_X.fit_transform(X)
y = sc_y.fit_transform(y)
```

Import the standard scaler function, which is helpful for generalizing numbers later aligned in plot visualizations to make the training data easier to use.

```
from sklearn.svm import SVR
regressor = SVR(kernel = 'rbf')
regressor.fit(X, y)
```

Import the support vector regression function with x and y data as reference in calculations

```
sc_y.inverse_transform(regressor.predict(sc_X.transform([[X]])))
```

Predicting fuel consumption by inputting variable data x.

```
X_grid = np.arange(min(sc_X.inverse_transform(X)), max(sc_X.inverse_transform(X)), 0.1)
X_grid = X_grid.reshape((len(X_grid), 1))
plt.scatter(sc_X.inverse_transform(X), sc_y.inverse_transform(y), color = 'red')
plt.plot(X_grid, sc_y.inverse_transform(regressor.predict(sc_X.transform(X_grid))), color = 'blue')
plt.title('Support Vector Regression')
plt.xlabel('4 Variables')
plt.ylabel('Fuel Consumption')
plt.show()
```

Plotting data testing for graphic visualization where the fuel consumption variable is the Y-axis and the other four variables are the X-axis. And the regression line will follow the calculation pattern

